Understanding the Counterexample Construction

Guided proof exercises

178 steps in twelve modules

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Purpose. Reconstruct the two main proofs in L. Cao's *Hierarchical Laminates Cannot Reproduce All Effective Properties of Planar Two-Phase Composites: A Counterexample with a 16-Component Potential* [1]. The exercises lead to Theorems 1.1 and 6.1 of the manuscript.

How to use the pair. Read a module's setup, then work in order. Use results from earlier steps when cited. Keep the answer book closed until you have tried a step; afterward compare the reasoning, not just the last formula. The answer book uses the same step numbers and repeats every question, so it can also serve as a self-contained guided reading of the proof. Three to five steps make a useful study session.

Difficulty key. Every question and its answer carry the same colored text label. The rating measures the work in that step, using the module's prerequisites and supplied background facts.

D1: Direct	A definition, calculation, or immediate application.
D2: Short proof	A short proof combining a few stated facts.
D3: Guided technical	A more delicate transition, with an explicit route or substeps in the question and a detailed answer.

The words and D1–D3 codes remain meaningful in grayscale. The definition finder, per-step lookup links, and background index are available in both books.

Prerequisites. Multivariable calculus, elementary real analysis, linear algebra including symmetric matrices, basic Hilbert spaces and weak derivatives, and introductory Fourier series. The supplied facts below cover the substantial general theory used by the manuscript. Prove their applications, rather than reproving the general theorems.

Background supplied as an input	Used in
Periodic Sobolev facts and Lax–Milgram (B1, B2)	Module 1
Separated-scale composition (B4)	Modules 2, 9
Fourier completeness and Parseval (B5, B6)	Modules 6, 10
The Cauchy determinant identity (B7)	Module 7
Weak derivatives and interface gluing (B8)	Module 8
Uniform higher integrability (B9)	Module 9
Smoothing and quasiconvex testing (B1, B10)	Module 12

Conventions used throughout. The cell is $Y = (0, 2\pi)^2$ and $\langle f \rangle = |Y|^{-1} \int_Y f$. A gradient is a $2 \times m$ matrix, with spatial derivatives in its rows. The vectors F_x, F_y are the transposed rows. The dot is the Frobenius inner product; $\|A\|$ for a matrix is its operator norm, whereas $|F|$ is the Frobenius norm of a gradient. $A \preceq B$ means $v \cdot Av \leq v \cdot Bv$ for all v. Energies have no factor $1/2$. Horizontal layers have normal $(0, 1)$. Constants in $O_\rho(N^{-1})$ may depend on the fixed contrast ρ but are independent of N.

Reading map

Module	Steps	Manuscript
1	1.1–1.12	Section 2.1
2	2.1–2.12	Sections 2.2–2.3, 3.1
3	3.1–3.14	Appendix A.1
4	4.1–4.12	Sections 3.1–3.3
5	5.1–5.17	Section 3.3, Appendix A.2
6	6.1–6.22	Sections 4.1–4.2, Appendix B
7	7.1–7.23	Section 4.3
8	8.1–8.18	Sections 5.1–5.2
9	9.1–9.12	Section 5.3
10	10.1–10.13	Section 6.1, planar duality
11	11.1–11.11	Section 6.2
12	12.1–12.12	Section 6.3, synthesis

The first theorem follows from Modules 1–9. The second theorem uses that binary witness and Modules 10–12. The essential chain is

Explicit matrix identities and a compact constitutive family
$\Downarrow$
An energy equality for every hierarchical laminate and its limits
$\Downarrow$
A convergent Fourier calculation gives a positive fifth-order difference
$\Downarrow$
A flux correction transfers it to finite binary stripes
$\Downarrow$
Exact prescribed fractions and an actual periodic witness
$\Downarrow$
Complementary energy and a rank-one convex minorant
$\Downarrow$
A strict gap between the two envelopes

Module 2 uses the composition theorem for the G-closure. Module 9 derives its exact-fraction correction from separated-scale recovery and the higher-integrability estimate.

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Notation and definition finder

The lookup line below each step links to its module guide and setup paragraphs. The "Uses" list links to earlier results needed for the argument. The table below locates the main definitions.

Notation	Meaning and first useful location
$Y, \langle f \rangle, \|f\|_p$	Periodic cell and normalized integral/norm; Setup 1.1; normalized L^p in Setup 9.2.
$H^1_{\mathrm{per}}, \mathcal{H}$	Periodic Sobolev space and its zero-mean subspace; Setups 1.1–1.2, B1.
$E \cdot J, \|E\|, \|A\|$	Frobenius product/norm for gradients and operator norm for tensors; Setup 1.2.
$A \preceq B$	Order of symmetric quadratic forms; Setup 1.2, B3.
u_F, e_F, j_F	Cell minimizer, $e_F = F + \nabla u_F$, and $j_F = L e_F$; Step 1.4–Step 1.6.
$B = L_*$	Effective tensor: $BF = \langle j_F \rangle$ and $F \cdot BF = \min\langle (F+\nabla v) \cdot L(F+\nabla v) \rangle$; Step 1.9–Step 1.10.
n, λ, z, F_1, F_2	Layer normal, first-material layer weight, compatible jump, and layer gradients; Step 2.1–Step 2.3.
$\chi, \theta, G_\theta, G_\theta^{\mathrm{L}}$	Binary indicator, its mean, and the two exact-fraction tensor closures; Setup 2.3, Step 2.11–Step 2.12.
O_j, Γ_j, Ω	Explicit 4×4 blocks, 8×8 generators, and $\Omega = \Gamma_1\Gamma_2\Gamma_3\Gamma_4$; Step 3.1–Step 3.6.
$Q(h), \mathcal{S}$	$Q(h) = \sum_j h_j \Gamma_j$ and $\mathcal{S} = \mathrm{span}\{I_8, \Gamma_1, \dots, \Gamma_4\}$; Setup 3.3, Step 3.7–Step 3.9.
P_j, P, S	Four row blocks, their stack $P \in \mathbb{R}^{16\times 8}$, and $S = P^\top P$; Step 3.11–Step 3.14.
$D, L(H), \mathcal{K}, \rho$	Constant vertical block, constitutive family, compact parameter family, and contrast; Setup 4.1, Step 4.2–Step 4.4.
$Q = PS^{-1}, \xi, \eta, \ell$	Lifting (a different Q from $Q(h)$), two detecting loads, and their energy difference; Step 4.6–Step 4.8.
$\mathcal{R}(k), T(k)$	$\mathcal{R}(k) = (k_x^2 I_{16} + k_y^2 D)^{-1}$ and its compressed multiplier $T(k)$; Step 4.9–Step 4.12. The convention at $k = 0$ is stated in Module 6.
$\widehat{f}(k), \|f\|_{\mathcal{A}}, \mathcal{T}$	Fourier coefficients, absolute-coefficient norm, and multiplier operator; Setups 6.1–6.2, Step 6.2–Step 6.5.
t, a, b, c, X, X_p, H_*	Smooth fraction field, its three frequencies, source, series terms, and mean source; Step 6.1–Step 6.8.
$d(M), K_j, \kappa$	Entry detector, partial frequency sums, and positive fifth-order constant; Step 7.1, Step 7.9, Step 7.20.
χ_N, h, t_k, w_N, r_N	Binary stripes, strip width, sampled fraction, zero-edge primitive, and constitutive residual; Step 8.1–Step 8.3, Step 8.11.
C_θ, B_χ, g_χ	Ideal tensor at fraction θ, selected actual periodic tensor, and $g_\chi = \ell(B_\chi) > 0$; Step 9.1–Step 9.3, Step 9.9.
$R_\perp, C^{\mathrm{d}}$	Row rotation and planar dual $R_\perp C^{-1} R_\perp^\top$; Setup 10.1, Step 10.2.

Notation	Meaning and first useful location
$F = (f, g), q_i, W$	Paired gradients, two quadratic branches, and their minimum; Setup 10.1, Step 10.9–Step 10.12.
$E_C, V, \mathrm{RC}\, W$	Paired energy, its minimum over $\mathcal{K}$, and rank-one convex envelope; Setup 11.1, Step 11.5–Step 11.8.
$\mathrm{QC}\, W, p, \overline{F}$	Quasiconvex envelope, $p = B_\chi \eta$, and mean paired gradient $(\xi, R_\perp p)$; Setup 12.1, B10.

Overloaded letters. A subscript $*$ denotes an effective object; H_* is an 8×8 mean source, whereas L_* is a tensor acting on gradients. In Module 9, A, B temporarily denote coefficient fields. In Step 12.3 they denote gradient matrices. Every such local reuse is specified in its setup. D always denotes the constant vertical block.

Background references. The keys B1–B12 link to the background index at the back, with precise hypotheses, use locations, and textbook or paper sections. The bibliography links to publisher or journal records where available.

1 The periodic cell problem, one step at a time

Prerequisites. Basic multivariable calculus and finite-dimensional inner products; no earlier course steps.

Definition lookup. The next two setup paragraphs define the periodic space, norms, weak equilibrium, corrector, and cell data. The effective tensor is defined in Step 1.9–Step 1.10.

Supplied background. B1, B2, B3. Each key links to its statement, scope, and reference.

Setup 1.1. Work on $Y = (0, 2\pi)^2$. Averages and L^2 norms are normalized: $\langle f \rangle = |Y|^{-1} \int_Y f$ and $\|f\|_2^2 = \langle |f|^2 \rangle$. A potential $v : Y \to \mathbb{R}^m$ has spatial-first gradient $\nabla v \in \mathbb{R}^{2\times m}$, with entries $(\nabla v)_{i\alpha} = \partial_i v_\alpha$. The dot denotes the Frobenius product. Let L be a measurable periodic self-adjoint coefficient field with $\mu I \preceq L(x) \preceq MI$ for almost every x, where $0 < \mu \le M$. Let $\mathcal{H} = \{v \in H^1_{\mathrm{per}}(Y; \mathbb{R}^m) : \langle v \rangle = 0\}$, $a(u, v) = \langle \nabla v \cdot L \nabla u \rangle$, and $b_F(v) = -\langle \nabla v \cdot LF \rangle$ for periodic H^1 arguments u, v and a constant mean gradient $F \in \mathbb{R}^{2\times m}$.

Setup 1.2. Definitions and background: H^1 consists of square-integrable functions with square-integrable weak first derivatives, and H^1_{per} is its closed subspace with matching opposite periodic traces (equivalently, the H^1 closure of smooth periodic functions). We use $\|v\|_{H^1}^2 = \|v\|_2^2 + \|\nabla v\|_2^2$. A Hilbert space is a complete inner-product space. The space H^1 with this norm is Hilbert, and its closed linear subspaces are Hilbert (B1). The Frobenius product is $E \cdot J = \sum_{i,\alpha} E_{i\alpha} J_{i\alpha}$; $\|A\| = \sup_{|z|=1} |Az|$ is the operator norm. Self-adjoint means $E \cdot LJ = J \cdot LE$, and $A \preceq B$ means $z \cdot Az \le z \cdot Bz$ for every z. Columnwise weak equilibrium $\operatorname{div} j = 0$ means $\langle \nabla \varphi \cdot j \rangle = 0$ for every smooth periodic vector test φ. A compatible field with mean F is $F + \nabla v$ with $v \in H^1_{\mathrm{per}}$. Smooth periodic density and the zero-mean Poincaré inequality $\|v\|_2 \le C_P \|\nabla v\|_2$ are supplied (B1). Lax–Milgram supplies a unique u solving $a(u, v) = b(v)$ on a Hilbert space when a is bounded and coercive and b is bounded linear (B2); coercive means $a(v, v) \ge c\|v\|_{H^1}^2$ for a constant $c > 0$. These theorems may be used throughout the exercises.

D2: Short proof

Step 1.1. The mean of a periodic gradient

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* local setup only.

Show that $\langle \nabla v \rangle = 0$ for every $v \in H^1_{\mathrm{per}}(Y; \mathbb{R}^m)$. First treat smooth v by integration, then use periodic smooth density and $|\langle g \rangle| \le \|g\|_2$ to pass to H^1.

D1: Direct

Step 1.2. Coercivity

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* local setup only.

Using the stated Poincaré inequality, find $c > 0$ such that $a(v, v) \ge c\|v\|_{H^1}^2$ on $\mathcal{H}$.

D1: Direct

Step 1.3. Bounded data

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* local setup only.

For $u, v \in \mathcal{H}$, show that $|b_F(v)| \le M|F|\|\nabla v\|_2$ and $|a(u, v)| \le M\|\nabla u\|_2 \|\nabla v\|_2$.

D2: Short proof

Step 1.4. Existence of the corrector

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.2, 1.3.

Apply Lax–Milgram to obtain a unique $u_F \in \mathcal{H}$ with $a(u_F, v) = b_F(v)$ for all $v \in \mathcal{H}$. Explain why the same equation holds for arbitrary periodic H^1 tests. For the Hilbert-space hypothesis, verify that $v \mapsto \langle v \rangle$ is continuous and that its zero set is closed.

D2: Short proof

Step 1.5. A weak solution minimizes energy

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4.

For $v \in H^1_{\mathrm{per}}(Y;\mathbb{R}^m)$, put $E_F(v) = \langle (F+\nabla v)\cdot L(F+\nabla v)\rangle$. Show that $E_F(v) - E_F(u_F) = a(v-u_F, v-u_F)$, and deduce uniqueness of the minimizing gradient. Write $w = v - u_F$, expand the quadratic, and test the weak equation with w.

D1: Direct

Step 1.6. Equilibrium

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4.

Set $e_F = F + \nabla u_F$ and $j_F = Le_F$. Show that $\operatorname{div} j_F = 0$ in distributions, where divergence is taken in each column.

D2: Short proof

Step 1.7. Linearity in the mean gradient

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4.

Use uniqueness in Step 1.4 to show $u_{\alpha F+\beta G} = \alpha u_F + \beta u_G$ for all constant $F, G \in \mathbb{R}^{2\times m}$ and real α, β.

D1: Direct

Step 1.8. The mean gradient really is prescribed

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.1, 1.6.

Show that $\langle e_F\rangle = F$.

D2: Short proof

Step 1.9. Symmetry of the mean-flux map

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4, 1.6, 1.7.

Define $BF = \langle j_F\rangle$, which is linear by Step 1.7. Show that $G\cdot BF = \langle e_G \cdot Le_F\rangle = F\cdot BG$ for all constant $F, G \in \mathbb{R}^{2\times m}$, hence B is self-adjoint. Use u_G as a test for u_F and u_F as a test for u_G.

D1: Direct

Step 1.10. The cell energy

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.4, 1.5, 1.9.

Show that $E_F(u_F) = F\cdot BF$. Thus the mean-flux definition and the minimum-energy definition give the same effective tensor $B = L_*$. Test the weak equation with u_F to remove the fluctuating part of the energy.

D2: Short proof

Step 1.11. Effective ellipticity

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* 1.1, 1.5, 1.10.

Show that $\mu|F|^2 \le F\cdot BF \le M|F|^2$. For the upper bound, try $v = 0$; for the lower bound, use $\langle \nabla v\rangle = 0$.

D2: Short proof

Step 1.12. A useful gradient–flux identity

Lookup: module 1 guide; setups 1.1, 1.2. *Uses:* local setup only.

If $j \in L^2_{\mathrm{per}}(Y;\mathbb{R}^{2\times m})$ is divergence-free and $\langle j\rangle = p$, show that $\langle (F+\nabla v)\cdot j\rangle = F\cdot p$ for every constant $F \in \mathbb{R}^{2\times m}$ and every periodic H^1 potential v. Extend the weak equilibrium identity from smooth tests by density.

2 Layers, the common block, and the two closures

Prerequisites. Cell minimization and testing: Step 1.4–Step 1.5, Step 1.10–Step 1.12.

Definition lookup. Layer coordinates and compatible fields are defined below; common spatial blocks are defined before Step 2.8; both closures and phase fraction are defined before Step 2.11.

Supplied background. B1, B4, B8. Each key links to its statement, scope, and reference.

Setup 2.1. Let B_1, B_2 be constant self-adjoint positive-definite tensors on $\mathbb{R}^{2\times m}$; positive definite means $E \cdot B_i E > 0$ for $E \neq 0$. Two planar layers have unit normal $n \in \mathbb{R}^2$, proportions $\lambda, 1-\lambda$ with $0 < \lambda < 1$, and mean gradient F. A unit tangent t satisfies $t \cdot n = 0$. The notation $n \otimes z$ means the matrix with entries $n_i z_\alpha$; the normal row $n^\top J$ is identified with a vector of $\mathbb{R}^m$. A compatible field means $F + \nabla v$ with periodic H^1 potential v, and weak equilibrium means $\langle \nabla\varphi \cdot j \rangle = 0$ for smooth periodic vector tests. For Steps 2.4–2.6 use the aligned period cell $\{sn + rt : 0 < s, r < 1\}$, with normalized averages and $s = n \cdot x$, $r = t \cdot x$. Module 1's variational proofs work on this cell as well. Periodicity on the aligned cell means invariance under translations by n and t. The resulting finite-dimensional minimum-energy formula defines the abstract laminate operation for every real unit normal (B4).

D1: Direct

Step 2.1. The compatible jump

Lookup: module 2 guide; setups 2.1. *Uses:* local setup only.

Continuity of a piecewise affine potential gives $t^\top(F_1 - F_2) = 0$. Show that $F_1 - F_2 = n \otimes z$ for some $z \in \mathbb{R}^m$.

D1: Direct

Step 2.2. The two layer gradients

Lookup: module 2 guide; setups 2.1. *Uses:* 2.1.

Solve $F_1 - F_2 = n \otimes z$ and $\lambda F_1 + (1-\lambda)F_2 = F$ to obtain $F_1 = F + (1-\lambda) n \otimes z$ and $F_2 = F - \lambda n \otimes z$.

D2: Short proof

Step 2.3. The normal-flux equation

Lookup: module 2 guide; setups 2.1. *Uses:* 2.2.

Let $\Phi(z) = \lambda F_1(z) \cdot B_1 F_1(z) + (1-\lambda) F_2(z) \cdot B_2 F_2(z)$, using Step 2.2. Differentiate in an arbitrary direction and show that its unique minimizer satisfies $n^\top B_1 F_1 = n^\top B_2 F_2$. To justify existence and uniqueness, define $K_i z = n^\top B_i (n \otimes z)$ and check that $(1-\lambda)K_1 + \lambda K_2$ is positive definite; the critical equation is a linear system with this matrix.

D2: Short proof

Step 2.4. Construct a compatible layer potential

Lookup: module 2 guide; setups 2.1. *Uses:* 2.2, 2.3.

For z from Step 2.3, construct a continuous periodic vector function $w(s)$ of period one with derivative $(1-\lambda)z$ on $0 < s < \lambda$ and $-\lambda z$ on $\lambda < s < 1$. Show that $F + \nabla[w(n \cdot x)]$ equals the two layer gradients on the aligned cell specified above. Start with $w(0) = 0$, integrate on each interval, and check the join at λ and the endpoint equality $w(1) = w(0)$.

D3: Guided technical

Step 2.5. Check equilibrium across the interfaces

Lookup: module 2 guide; setups 2.1. *Uses:* 2.3, 2.4.

Let $j = B_i F_i$ in layer i. Use the normal-flux equality from Step 2.3 to show that $\operatorname{div} j = 0$ in distributions, including across the periodic join. In the aligned coordinates $x = sn + rt$, write $j \cdot \nabla\varphi = (n^\top j) \cdot \partial_s\varphi + (t^\top j) \cdot \partial_r\varphi$. The normal flux $n^\top j$ is the same in both layers; the tangential flux $t^\top j$ is independent of r. Integrate the two terms over $0 < s, r < 1$.

D2: Short proof

Step 2.6. Prove global minimizing energy

Lookup: module 2 guide; setups 2.1. *Uses:* 1.12, 2.4, 2.5.

Let e be the compatible field from Step 2.4 and let $L = B_i$ in layer i. Using Step 2.5, show $\langle (e + \nabla v) \cdot L(e + \nabla v) \rangle \geq \langle e \cdot Le \rangle$ for every periodic H^1 potential v. Expand the quadratic and use Step 1.12 to cancel the cross term; explain why this compares with every compatible field of mean F.

D1: Direct

Step 2.7. The laminate energy bound

Lookup: module 2 guide; setups 2.1. *Uses:* 1.9, 1.10, 2.2, 2.6.

By Step 2.4–Step 2.6, the cell tensor of the layers satisfies $F \cdot B_{\mathrm{lam}} F = \min_z \Phi(z)$. Show $F \cdot B_{\mathrm{lam}} F \leq \lambda F \cdot B_1 F + (1-\lambda) F \cdot B_2 F$, and extend the definition by identifying B_{lam} at $\lambda = 0, 1$.

Setup 2.2. For Steps 2.8–2.10, assume the measurable periodic self-adjoint coefficient field L is uniformly elliptic, $\mu I \preceq L \preceq MI$ almost everywhere, and has spatial block form

$$L(x, y) = \operatorname{diag}(A(x, y), D).$$

Here $A(x, y), D \in \mathbb{R}^{m \times m}$, $0 < \mu \leq M$, and D is constant positive definite. Write the transposed spatial rows of a mean gradient as F_x, F_y.

D1: Direct

Step 2.8. A vertical mean needs no corrector

Lookup: module 2 guide; setups 2.1, 2.2. *Uses:* 1.1, 1.4.

For $F_x = 0$ and $F_y = q$ constant, show that $u_F = 0$ solves the weak cell equation. Determine the resulting mean flux.

D2: Short proof

Step 2.9. The vertical block survives homogenization

Lookup: module 2 guide; setups 2.1, 2.2. *Uses:* 1.1, 1.7, 1.9, 1.11, 2.8.

For a horizontal mean gradient F with $F_y = 0$, show that the mean vertical flux is zero. Combine this with Step 2.8 to obtain $L_* = \operatorname{diag}(A_*, D)$ for some positive-definite A_*.

D1: Direct

Step 2.10. Horizontal layers average arithmetically

Lookup: module 2 guide; setups 2.1, 2.2. *Uses:* 2.2, 2.3, 2.7.

For $B_i = \operatorname{diag}(A_i, D)$ and $n = (0, 1)$, show that $z = 0$ minimizes Φ and that $B_{\mathrm{lam}} = \lambda B_1 + (1-\lambda) B_2$.

Setup 2.3. Fix constant self-adjoint positive-definite original phases L_0, L_1 on $\mathbb{R}^{2\times m}$. A binary coefficient field is $L_\chi = (1 - \chi) L_0 + \chi L_1$, where the phase indicator $\chi : Y \to \{0, 1\}$ is measurable and periodic. Its phase-one fraction is the proportion $\theta = \langle \chi \rangle$ of the cell occupied by phase one. For $0 \leq \theta \leq 1$, let G_θ be the operator-norm closure of the effective tensors $(L_\chi)_*$ at exact fraction $\langle \chi \rangle = \theta$. A finite hierarchical laminate repeatedly layers earlier tensors, beginning with L_0, L_1, and records the original-phase fraction at every node. Let G_θ^{L} be the operator-norm closure of its

tensors at exact fraction θ. A norm closure consists of all limits in operator norm of sequences from the original set. The following composition theorem may be used: for $C_i \in G_{\theta_i}$ and $0 \le \lambda \le 1$, layering C_1, C_2 at proportions $\lambda, 1-\lambda$ gives a tensor in $G_{\lambda\theta_1+(1-\lambda)\theta_2}$, with exact-fraction periodic recovery (B4).

D1: Direct

Step 2.11. Fraction bookkeeping

Lookup: module 2 guide; setups 2.1, 2.2, 2.3. *Uses:* local setup only.

If the two materials in an outer laminate have phase-one fractions $\theta_1, \theta_2 \in [0,1]$ and outer proportions $\lambda, 1-\lambda$ with $0 \le \lambda \le 1$, compute the overall phase-one fraction.

D2: Short proof

Step 2.12. Why every finite laminate lies in the G-closure

Lookup: module 2 guide; setups 2.1, 2.2, 2.3. *Uses:* 2.11.

Use the composition theorem above and induction on the number of layering operations to show $G_\theta^{\mathrm{L}} \subseteq G_\theta$.

3 The explicit matrices and the eight auxiliary coordinates

Prerequisites. Matrix multiplication, transpose, and the real symmetric spectral theorem; no earlier course steps.

Definition lookup. The first setup defines O_j, Γ_j; $Q(h), \mathcal{S}$ are recorded before Step 3.7; P_j, P, S are recorded before Step 3.11. The lookup paragraph below recalls spectral and Gram terminology.

Supplied background. B3. Each key links to its statement, scope, and reference.

Setup 3.1. For $\beta = (\beta_0, \beta_1, \beta_2, \beta_3) \in \mathbb{R}^4$ define

$$O(\beta) = \begin{pmatrix} \beta_0 & -\beta_1 & -\beta_2 & -\beta_3 \\ \beta_1 & \beta_0 & -\beta_3 & \beta_2 \\ \beta_2 & \beta_3 & \beta_0 & -\beta_1 \\ \beta_3 & -\beta_2 & \beta_1 & \beta_0 \end{pmatrix}.$$

Set $O_1 = I_4$, $O_2 = O(0,1,0,0)$, $O_3 = O(0,0,1,0)$, $O_4 = O(0,0,0,-1)$, and $\Gamma_j = \begin{pmatrix} 0 & O_j \\ O_j^\top & 0 \end{pmatrix}$. Here $j \in \{1,2,3,4\}$, and all matrices are real.

Setup 3.2. Matrix lookup: I_d is the $d \times d$ identity and e_r is a coordinate vector. The anticommutator of A, B is $AB + BA$; δ_{ij} is 1 when $i = j$ and 0 otherwise. The spectral theorem says that a real symmetric matrix has an orthonormal eigenbasis; its trace, the sum of its diagonal entries, is also the sum of its eigenvalues with multiplicity. Its operator norm is the largest absolute eigenvalue, and $\sup_{|e|=1} e^\top A e$ is its largest eigenvalue. A Gram matrix is $P^\top P$, with $e^\top P^\top P e = |Pe|^2$; full column rank means $Pe = 0$ only for $e = 0$ (B3).

D1: Direct

Step 3.1. The three signed permutation matrices

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* local setup only.

Write O_2, O_3, O_4 explicitly from the displayed formula for $O(\beta)$.

D1: Direct

Step 3.2. Transpose and square

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.1.

Using the sparse matrices from Step 3.1, show $O_j^\top = -O_j$ and $O_j^2 = -I_4$ for $j = 2, 3, 4$. Deduce $O_j^\top O_j = I_4$.

D1: Direct

Step 3.3. One four-dimensional multiplication

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.2.

Verify $O_2 O_3 = -O_4$ by comparing the two sides on the four coordinate vectors.

D2: Short proof

Step 3.4. The remaining anticommutations

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.2, 3.3.

Use Steps 3.2–3.3 to show $O_i O_j = -O_j O_i$ for distinct $i, j \in \{2, 3, 4\}$. First transpose $O_2 O_3 = -O_4$, using $(AB)^\top = B^\top A^\top$; then substitute $O_4 = -O_2 O_3$ in the remaining two pairs.

D2: Short proof

Step 3.5. The Clifford relations

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.2, 3.4.

By multiplying 2×2 blocks, show that $\Gamma_j^\top = \Gamma_j$ and $\Gamma_i \Gamma_j + \Gamma_j \Gamma_i = 2\delta_{ij} I_8$. Treat separately $i = j$, one index equal to 1, and distinct indices both larger than 1.

D2: Short proof

Step 3.6. The sign of the four-factor product

Lookup: module 3 guide; setups 3.1, 3.2. *Uses:* 3.2, 3.3.

Use $O_2O_3 = -O_4$ to show $\Gamma_1\Gamma_2\Gamma_3\Gamma_4 = \operatorname{diag}(I_4, -I_4)$. Compute the two block-diagonal factors $\Gamma_1\Gamma_2$ and $\Gamma_3\Gamma_4$ before multiplying them.

Setup 3.3. Write $Q(h) = \sum_{j=1}^{4} h_j\Gamma_j$, $h \in \mathbb{R}^4$, and $\mathcal{S} = \{h_0 I_8 + Q(h) : h_0 \in \mathbb{R},\ h \in \mathbb{R}^4\}$.

D1: Direct

Step 3.7. Cancellation of the mixed terms

Lookup: module 3 guide; setups 3.1, 3.2, 3.3. *Uses:* 3.5.

Expand $Q(h)^2$ and prove $Q(h)^2 = |h|^2 I_8$.

D2: Short proof

Step 3.8. The spectrum

Lookup: module 3 guide; setups 3.1, 3.2, 3.3. *Uses:* 3.5, 3.7.

For $h \neq 0$, use Step 3.7 and $\operatorname{tr} Q(h) = 0$ to show that $h_0 I_8 + Q(h)$ has eigenvalues $h_0 \pm |h|$, each four times. State the spectrum for $h = 0$. First apply the symmetric spectral theorem to $Q(h)$; then determine the two multiplicities from their sum 8 and their weighted sum $\operatorname{tr} Q(h) = 0$.

D1: Direct

Step 3.9. The inverse stays in the same space

Lookup: module 3 guide; setups 3.1, 3.2, 3.3. *Uses:* 3.7, 3.8.

For invertible $H = h_0 I_8 + Q(h)$, show

$$H^{-1} = \frac{h_0 I_8 - Q(h)}{h_0^2 - |h|^2}.$$

Use Step 3.8 to justify that the denominator is nonzero.

D2: Short proof

Step 3.10. Inverses are different from products

Lookup: module 3 guide; setups 3.1, 3.2, 3.3. *Uses:* 3.5, 3.9.

Show that $\Gamma_1\Gamma_2$ is a nonzero skew-symmetric matrix and hence is not in $\mathcal{S}$. Conclude that $\mathcal{S}$ is not closed under ordinary multiplication.

Setup 3.4. Let $P_j = \frac{1}{2}(I_4 \;\; O_j) \in \mathbb{R}^{4\times 8}$, stack them to form $P \in \mathbb{R}^{16\times 8}$, and put $S = P^\top P$.

D1: Direct

Step 3.11. The four Gram factors

Lookup: module 3 guide; setups 3.1, 3.2, 3.3, 3.4. *Uses:* 3.2.

Show that $P_j^\top P_j = \frac{1}{4}(I_8 + \Gamma_j)$.

D1: Direct

Step 3.12. The Gram spectrum

Lookup: module 3 guide; setups 3.1, 3.2, 3.3, 3.4. *Uses:* 3.8, 3.11.

Sum Step 3.11 to obtain $S = I_8 + \frac{1}{4}\sum_j \Gamma_j$. Use Step 3.8 to show that its eigenvalues are $1/2$ and $3/2$, each four times.

D1: Direct

Step 3.13. An explicit inverse of the Gram matrix

Lookup: module 3 guide; setups 3.1, 3.2, 3.3, 3.4. *Uses:* 3.9, 3.12.

Use Step 3.9 to show $S^{-1} = \frac{4}{3} I_8 - \frac{1}{3}\sum_j \Gamma_j$.

D2: Short proof

Step 3.14. Rank and operator norm

Lookup: module 3 guide; setups 3.1, 3.2, 3.3, 3.4. *Uses:* 3.12.

Use $P^\top P = S$ and Step 3.12 to prove that P has full column rank and $\|P\|^2 = 3/2$. Use $|Pe|^2 = e^\top S e$ for injectivity, and maximize this expression over unit e for the norm.

4 The phases, the detecting loads, and the directional matrix

Prerequisites. Common-block layers: Step 2.10; auxiliary matrices, spectrum, and norm: Step 3.5, Step 3.11–Step 3.14.

Definition lookup. The first setup defines $L(H), \mathcal{K}, D, \rho$ and recalls $\mathcal{S}, P, S$; the setups before Step 4.7 and Step 4.9 define the loads, detector, $\mathcal{R}(k)$, and $T(k)$.

Supplied background. B3. Each key links to its statement, scope, and reference.

Setup 4.1. Use $P, S, \Gamma_j, \mathcal{S}$ from Module 3 and set $D = \operatorname{diag}(1, 1/2, 1/3, 1/4) \otimes I_4$. Fix $0 < \rho < 2/3$ and define

$$L(H) = \operatorname{diag}(I_{16} - PHP^\top, D), \qquad \mathcal{K} = \{L(H) : H \in \mathcal{S},\ 0 \preceq H \preceq \rho I_8\}.$$

The blocks act on the horizontal and vertical rows of a 2×16 gradient. The notation $A \preceq B$ means $v^\top A v \le v^\top B v$ for every vector v.

Setup 4.2. Auxiliary lookup: $\mathcal{S} = \operatorname{span}\{I_8, \Gamma_1, \dots, \Gamma_4\}$; each Γ_j is symmetric with zero diagonal 4×4 blocks. The matrices $P_j = \frac{1}{2}(I_4 \ \ O_j)$ are stacked to form $P \in \mathbb{R}^{16\times 8}$, and $S = P^\top P = I_8 + \frac{1}{4}\sum_j \Gamma_j$. The identities needed here are $\frac{1}{2} I_8 \preceq S \preceq \frac{3}{2} I_8$, $\|P\|^2 = 3/2$, and $S^{-1} = \frac{4}{3} I_8 - \frac{1}{3}\sum_j \Gamma_j$ (Step 3.11–Step 3.14). Here $\operatorname{diag}(A, D)$ acts separately on the horizontal and vertical spatial rows, $\otimes I_4$ repeats each listed scalar as a 4×4 block, and all matrix norms are operator norms.

D1: Direct

Step 4.1. The two diagonal blocks agree

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* local setup only.

If $H = h_0 I_8 + \sum_j h_j \Gamma_j$, show that both diagonal 4×4 blocks of H equal $h_0 I_4$.

D2: Short proof

Step 4.2. A uniform ellipticity bound

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* 3.14.

Show that every $C \in \mathcal{K}$ satisfies $\mu_\rho I \preceq C \preceq I$, where $\mu_\rho = \min\{1 - 3\rho/2, 1/4\} > 0$. Bound $f^\top PHP^\top f$ by $\rho |P^\top f|^2$ and use $\|P^\top\| = \|P\|$.

D2: Short proof

Step 4.3. Compactness

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* local setup only.

Show that $\{H \in \mathcal{S} : 0 \preceq H \preceq \rho I_8\}$ is closed and bounded, and deduce that $\mathcal{K}$ is compact. Pass each quadratic-form inequality to a matrix limit, use finite-dimensional closed-and-bounded compactness, and check continuity of $H \mapsto L(H)$.

D2: Short proof

Step 4.4. The two original phases

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* 2.10.

Define $L_s = L(\rho s I_8)$ for $0 \le s \le 1$. Show $L_s = (1 - s)L_0 + sL_1$ and $L_1 \preceq L_0$; identify the horizontal layering that realizes L_s.

D1: Direct

Step 4.5. Lifting an auxiliary vector

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* 3.12.

For $e \in \mathbb{R}^8$, show that $f = PS^{-1}e$ satisfies $P^\top f = e$.

D1: Direct

Step 4.6. The reduced energy matrix

Lookup: module 4 guide; setups 4.1, 4.2. *Uses:* 4.5.

For a symmetric $H \in \mathbb{R}^{8\times 8}$, let $Q = PS^{-1} \in \mathbb{R}^{16\times 8}$ (a lifting matrix, distinct from the map $h \mapsto Q(h)$ in Module 3). First compute $P^\top Q$ and $Q^\top Q$, then show $Q^\top(I_{16} - PHP^\top)Q = S^{-1} - H$.

Setup 4.3. Let $e_+ = e_1$ and $e_- = e_5$ be coordinate vectors of $\mathbb{R}^8$. Define horizontal mean gradients ξ, η by $\xi_x = PS^{-1}e_+$, $\eta_x = PS^{-1}e_-$, and $\xi_y = \eta_y = 0$. Define $\ell(B) = \eta \cdot B\eta - \xi \cdot B\xi$.

D2: Short proof

Step 4.7. Equal energies at two distinct loads

Lookup: module 4 guide; setups 4.1, 4.2, 4.3. *Uses:* 3.13, 4.1, 4.6.

For $H = h_0 I_8 + \sum_j h_j \Gamma_j$ with $L(H) \in \mathcal{K}$, use Steps 4.1 and 4.6 to show $\xi \cdot L(H)\xi = \eta \cdot L(H)\eta = 4/3 - h_0$. Deduce $|\xi|^2 = |\eta|^2 = 4/3$ and check that the loads are distinct. Use Step 3.13 to read the two needed diagonal entries of S^{-1}, and use $Q^\top Q = S^{-1}$ for the norms.

D1: Direct

Step 4.8. Continuity of the detector

Lookup: module 4 guide; setups 4.1, 4.2, 4.3. *Uses:* 4.7.

For linear maps B, C on $\mathbb{R}^{2\times 16}$, show $|\ell(B) - \ell(C)| \le (8/3)\|B - C\|$. In particular, $\ell = 0$ on every norm limit of tensors on which it vanishes.

Setup 4.4. For nonzero $k = (k_x, k_y) \in \mathbb{R}^2$, define $\mathcal{R}(k) = (k_x^2 I_{16} + k_y^2 D)^{-1}$ and $T(k) = k_x^2 P^\top \mathcal{R}(k) P$. The inverse exists because D is positive definite and $k \neq 0$.

D1: Direct

Step 4.9. The cost of a normal gradient

Lookup: module 4 guide; setups 4.1, 4.2, 4.3, 4.4. *Uses:* local setup only.

For a unit normal $n = (n_x, n_y) \in \mathbb{R}^2$, a vector $z \in \mathbb{R}^{16}$, and the baseline phase $L_0 = \mathrm{diag}(I_{16}, D)$, show $(n \otimes z) \cdot L_0(n \otimes z) = z^\top(n_x^2 I_{16} + n_y^2 D)z$.

D2: Short proof

Step 4.10. The four directional weights

Lookup: module 4 guide; setups 4.1, 4.2, 4.3, 4.4. *Uses:* 3.11.

Use the four diagonal blocks of D and Step 3.11 to derive

$$T(k) = \frac{1}{4}\sum_{j=1}^{4} \frac{j k_x^2}{j k_x^2 + k_y^2}(I_8 + \Gamma_j).$$

D2: Short proof

Step 4.11. A uniform bound on the directional matrix

Lookup: module 4 guide; setups 4.1, 4.2, 4.3, 4.4. *Uses:* 3.11, 3.12, 4.10.

Using weights between zero and one in Step 4.10, show $T(k) \in \mathcal{S}$ and $0 \preceq T(k) \preceq S \preceq (3/2)I_8$. For the order bound, rewrite the sum as $\sum_j w_j P_j^\top P_j$ with $0 \le w_j \le 1$; use the positive semidefinite Gram factors $P_j^\top P_j$.

D2: Short proof

Step 4.12. Which directions can be inverted?

Lookup: module 4 guide; setups 4.1, 4.2, 4.3, 4.4. *Uses:* 3.14.

Show that $T(k) = 0$ if $k_x = 0$, whereas $T(k)$ is positive definite if $k_x \neq 0$. Also observe directly from its definition that $T(-k) = T(k)$ and $T(ak) = T(k)$ for every nonzero real a.

5 One layering step and the invariant

Prerequisites. The layer minimum: Step 2.2–Step 2.7; inverse closure: Step 3.9; phase ellipticity and directional matrices: Step 4.2, Step 4.11–Step 4.12.

Definition lookup. The setup below defines $L(H), \mathcal{K}, \overline{H}, \widetilde{H}, \mathcal{R}(n), T(n)$ and the layer fields. The lookup paragraph recalls matrix order and defines H_{lam}; Step 5.7 and Step 5.10 introduce the shifted matrices.

Supplied background. B3, B4. Each key links to its statement, scope, and reference.

Setup 5.1. Use the matrices P, D, Γ_j from Modules 3–4 and put $L(H) = \mathrm{diag}(I_{16} - PHP^\top, D)$. Thus $P \in \mathbb{R}^{16\times 8}$ has full column rank, $D \succeq I_{16}/4$ is constant, and $\|P\|^2 = 3/2$. The space $\mathcal{S} = \mathrm{span}\{I_8, \Gamma_1, \dots, \Gamma_4\}$ consists of real symmetric matrices and is linear and closed under inversion of its invertible members. Define $\mathcal{K} = \{L(A) : A \in \mathcal{S},\ 0 \preceq A \preceq \rho I_8\}$. Fix $0 < \rho < 2/3$ and $H^{(1)}, H^{(2)} \in \mathcal{S}$ with $0 \preceq H^{(i)} \preceq \rho I_8$. Thus $L(H^{(i)}) \succeq \mu I$ for $\mu = \min\{1/4, 1 - 3\rho/2\} > 0$.

Let $n = (n_x, n_y)$ be a unit normal and $0 < \lambda < 1$. Set $\Delta H = H^{(1)} - H^{(2)}$, $\overline{H} = \lambda H^{(1)} + (1-\lambda)H^{(2)}$, and $\widetilde{H} = (1-\lambda)H^{(1)} + \lambda H^{(2)}$. For $F \in \mathbb{R}^{2\times 16}$ and $z \in \mathbb{R}^{16}$, the compatible layer gradients are $F_1 = F + (1-\lambda)n \otimes z$ and $F_2 = F - \lambda n \otimes z$. Their weighted energy is $\mathcal{E}_F(z) = \lambda F_1 \cdot L(H^{(1)})F_1 + (1-\lambda)F_2 \cdot L(H^{(2)})F_2$.

By Step 4.11–Step 4.12, $\mathcal{R}(n) = (n_x^2 I_{16} + n_y^2 D)^{-1}$ and $T(n) = n_x^2 P^\top \mathcal{R}(n) P \in \mathcal{S}$, with $0 \preceq T(n) \preceq (3/2)I_8$. It is positive definite when $n_x \neq 0$ and zero when $n_x = 0$. Matrix factors are multiplied in their displayed order. References: manuscript Section 3.3 and Appendix A.2.

Setup 5.2. Lookup: $A \preceq B$ means $v^\top(B-A)v \geq 0$ for every vector v; $A \succ 0$ means the inequality $v^\top Av > 0$ for every nonzero v. All order comparisons here are between real symmetric matrices. For $A \succ 0$, the inverse square root $A^{-1/2}$ is obtained by replacing each eigenvalue $a > 0$ with $a^{-1/2}$ in an orthogonal diagonalization of A. The operator norm is $\|A\| = \sup_{|v|=1} |Av|$, and $\mathrm{diag}(A, D)$ acts separately on the transposed spatial rows F_x, F_y. The matrix $n \otimes z$ has rows $n_x z, n_y z$. By Step 2.7, $\min_z \mathcal{E}_F(z) = F \cdot L_{\mathrm{lam}} F$ is the minimum energy over all compatible periodic fields on the aligned layer cell of Module 2. The calculation in Step 5.5 identifies $L_{\mathrm{lam}} = L(H_{\mathrm{lam}})$ and thereby defines the parameter H_{lam}. A finite hierarchical laminate is a finite binary tree of these layering operations, with the constituent phases at its leaves.

D1: Direct

Step 5.1. Check the layer mean

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* local setup only.

Show that $\lambda F_1 + (1-\lambda)F_2 = F$ and $F_1 - F_2 = n \otimes z$.

D1: Direct

Step 5.2. Find the constant energy term

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* local setup only.

Evaluate $\mathcal{E}_F(0)$ and show that it equals $F \cdot L(\overline{H})F$.

D1: Direct

Step 5.3. Find the linear term

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* local setup only.

Expand only the terms linear in z in $\mathcal{E}_F(z)$. Show that their sum is

$$-2\lambda(1-\lambda)n_x z^\top P \Delta H P^\top F_x.$$

D2: Short proof

Step 5.4. Find and bound the quadratic matrix

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* 4.2.

Show that the quadratic term is $\lambda(1-\lambda)z^\top Mz$, where $M = n_x^2 I_{16} + n_y^2 D - n_x^2 P\widetilde{H}P^\top$. Prove $M \succeq \mu I_{16}$.

D2: Short proof

Step 5.5. Complete one quadratic square

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* 2.7, 5.2, 5.3, 5.4.

Using steps 5.2–5.4, complete the square in z and identify the minimum energy as $F \cdot L(H_{\mathrm{lam}})F$, where

$$H_{\mathrm{lam}} = \overline{H} + \lambda(1-\lambda)\Delta H[n_x^2 P^\top M^{-1}P]\Delta H.$$

First put $b = n_x P\Delta H P^\top F_x$; the z-dependent part is $\lambda(1-\lambda)(z^\top Mz - 2z^\top b)$.

D1: Direct

Step 5.6. Handle horizontal layers separately

Lookup: module 5 guide; setups 5.1, 5.2. *Uses:* 5.5.

Set $n_x = 0$ in step 5.5. Show $H_{\mathrm{lam}} = \overline{H}$, and explain why no inverse of $T(n)$ is needed in this case.

Setup 5.3. For Steps 5.7–5.12 assume $n_x \neq 0$, so $T(n)$ is invertible.

D2: Short proof

Step 5.7. Check the shifted inverses

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 3.9, 4.11, 4.12.

Assume $n_x \neq 0$ and set $\Lambda = T(n)^{-1}$. Show that $\Lambda \in \mathcal{S}$, $\Lambda \succeq (2/3)I_8$, and $\Lambda - H^{(i)}, \Lambda - \widetilde{H} \succ 0$.

D1: Direct

Step 5.8. Verify the reduction identity

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.4.

Show that $M\mathcal{R}(n)P = P[I_8 - \widetilde{H}T(n)]$ by multiplying out the left side.

D2: Short proof

Step 5.9. Reduce the inverse correction

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.4, 5.7, 5.8.

Use step 5.8 and $I_8 - \widetilde{H}T(n) = (\Lambda - \widetilde{H})T(n)$ to prove $n_x^2 P^\top M^{-1}P = (\Lambda - \widetilde{H})^{-1}$. Set $J = I_8 - \widetilde{H}T(n)$ and solve the identity in step 5.8 for $M^{-1}P$; use $(AB)^{-1} = B^{-1}A^{-1}$.

D1: Direct

Step 5.10. Factor the weighted inverse sum

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.7.

Set $K_i = \Lambda - H^{(i)}$ and $K_\lambda = (1-\lambda)K_1 + \lambda K_2$. Prove $\lambda K_1^{-1} + (1-\lambda)K_2^{-1} = K_2^{-1}K_\lambda K_1^{-1}$.

D1: Direct

Step 5.11. Expand the harmonic mean in small steps

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.7, 5.10.

Let $\Delta K = K_1 - K_2$. Substitute $K_1 = K_\lambda + \lambda\Delta K$ and $K_2 = K_\lambda - (1-\lambda)\Delta K$ into $K_1 K_\lambda^{-1} K_2$, and simplify.

D2: Short proof

Step 5.12. Obtain the shifted harmonic formula

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.5, 5.9, 5.10, 5.11.

Combine steps 5.5 and 5.9–5.11 to show $H_{\mathrm{lam}} = \Lambda - [\lambda K_1^{-1} + (1-\lambda)K_2^{-1}]^{-1}$.

D2: Short proof

Step 5.13. Preserve the matrix space

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 3.9, 5.6, 5.7, 5.10, 5.12.

Use the formula in step 5.12 to show $H_{\mathrm{lam}} \in \mathcal{S}$. For $n_x = 0$, use step 5.6 instead.

D2: Short proof

Step 5.14. Prove reversal of matrix order

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* local setup only.

If $0 \prec A \preceq B$, show $B^{-1} \preceq A^{-1}$. Hint: diagonalize $A^{-1/2}BA^{-1/2}$.

D2: Short proof

Step 5.15. Preserve the order interval

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.6, 5.7, 5.10, 5.12, 5.14.

For $n_x \neq 0$, start from $\Lambda - \rho I_8 \preceq K_i \preceq \Lambda$ and use step 5.14 twice to prove $0 \preceq H_{\mathrm{lam}} \preceq \rho I_8$. For $n_x = 0$, use step 5.6.

D2: Short proof

Step 5.16. Induct over a finite hierarchy

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.13, 5.15.

Assume $L_0, L_1 \in \mathcal{K}$. Use preservation of $\mathcal{K}$ under one layering step to prove that every finite hierarchical laminate belongs to $\mathcal{K}$.

D1: Direct

Step 5.17. Pass to tensor limits

Lookup: module 5 guide; setups 5.1, 5.2, 5.3. *Uses:* 5.16.

Assume $\mathcal{K}$ is compact and a continuous functional ℓ vanishes on $\mathcal{K}$. Show that every limit of finite laminate tensors of L_0, L_1 satisfies $\ell = 0$, even if their phase fractions vary.

6 A convergent Fourier cell calculation

Prerequisites. The weak cell problem and mean flux: Step 1.5–Step 1.10; common vertical block: Step 2.8–Step 2.9; matrix rank and ellipticity: Step 3.14, Step 4.2; directional bound: Step 4.11–Step 4.12.

Definition lookup. The setup below defines $t, a, b, c, H, \mathcal{R}, T, \mathcal{T}$ and the Wiener norm; its lookup paragraph defines Fourier support, weak derivatives, and H^1_{per}. The series and averaged parameter are defined in Step 6.6–Step 6.8; σ_ζ is defined in Step 6.20.

Supplied background. B3, B4, B5, B6, B12. Each key links to its statement, scope, and reference.

Setup 6.1. Work on $Y = (0, 2\pi)^2$ with normalized averages. Define

$$t(x,y) = \frac{1}{2} + \frac{\cos x + \cos y + \cos(x+3y)}{6}, \qquad a = (1,0), \quad b = (0,1), \quad c = (1,3).$$

Use $H = \rho t I_8$, $0 < \rho < 2/3$, and $\overline{L} = L(H) = \operatorname{diag}(I_{16} - PHP^\top, D)$. Here $\mathcal{K} = \{L(A) : A \in \mathcal{S},\ 0 \preceq A \preceq \rho I_8\}$ and $\mathcal{S} = \operatorname{span}\{I_8, \Gamma_1, \dots, \Gamma_4\}$. The matrices P, D, Γ_j are fixed as in Modules 3–4: $D \succeq I_{16}/4$ is constant, P has full column rank, and $\|P\|^2 = 3/2$. For $k \neq 0$, set $\mathcal{R}(k) = (k_x^2 I_{16} + k_y^2 D)^{-1}$ and $T(k) = k_x^2 P^\top \mathcal{R}(k) P$; set $T(0) = 0$. Module 4 gives $T(-k) = T(k)$, $\|T(k)\| \le 3/2$, and $T(b) = 0$.

Our Fourier convention is

$$\widehat{M}(k) = \langle M(x,y) e^{-\mathrm{i}k\cdot(x,y)} \rangle, \qquad \widehat{\partial_\nu M}(k) = \mathrm{i}k_\nu \widehat{M}(k).$$

Define $\widehat{\mathcal{T}M}(k) = T(k)\widehat{M}(k)$ and $\|M\|_{\mathcal{A}} = \sum_{k\in\mathbb{Z}^2} \|\widehat{M}(k)\|$, using the matrix operator norm. Completeness of the space of absolutely summable coefficient sequences is supplied; it makes this Wiener space complete. A horizontal mean gradient ζ means $\zeta_y = 0$, with transposed first row $\zeta_x \in \mathbb{R}^{16}$. References: manuscript Sections 4.1–4.2 and Appendix B.

Setup 6.2. Fourier lookup: $\langle M \rangle = (2\pi)^{-2} \int_Y M$; the exponentials have $\langle e^{\mathrm{i}(k-l)\cdot(x,y)} \rangle = 1$ if $k = l$ and 0 otherwise. The Fourier support is the set of k with $\widehat{M}(k) \neq 0$. A Fourier multiplier acts on each coefficient separately and creates no new frequencies. The Wiener space $\mathcal{A}$ consists of absolutely summable coefficients, identified with their Fourier series as justified in Step 6.3. Its matrix norm also applies to complex coefficients, with the Euclidean norm on complex vectors. The same Wiener and supremum notation applies to scalar fields using absolute value and to vector fields using the Euclidean norm. Here $\|M\|_\infty = \sup_{(x,y)\in Y} \|M(x,y)\|$ for continuous fields. A weak derivative $g = \partial_\nu u$ satisfies $\langle g \cdot \varphi \rangle = -\langle u \cdot \partial_\nu \varphi \rangle$ for every smooth periodic test φ. The space H^1_{per} consists of periodic L^2 functions with both weak first derivatives in L^2. A corrector is the zero-mean weak solution from Step 1.4; its mean flux defines the effective tensor by Step 1.9.

D1: Direct

Step 6.1. Bound the local fraction

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* local setup only.

Show $0 \le t \le 1$ and $\langle t \rangle = 1/2$. Deduce that $\overline{L}(x,y)$ takes values in $\mathcal{K}$.

D1: Direct

Step 6.2. List the Fourier data

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* local setup only.

Compute $\widehat{t}(k)$ and show $\|t\|_{\mathcal{A}} = 1$.

D2: Short proof

Step 6.3. Control a function by its coefficients

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* local setup only.

Show $\|M\|_\infty \le \|M\|_{\mathcal{A}}$. Explain why the Fourier series converges uniformly when the right side is finite.

D2: Short proof

Step 6.4. Bound one product

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.3.

Use $\widehat{MN}(k) = \sum_l \widehat{M}(l)\widehat{N}(k-l)$ to prove $\|MN\|_{\mathcal{A}} \le \|M\|_{\mathcal{A}}\|N\|_{\mathcal{A}}$.

D1: Direct

Step 6.5. Bound the multiplier

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 1.1, 4.11.

Use $\|T(k)\| \le 3/2$ to show $\|\mathcal{T}M\|_{\mathcal{A}} \le (3/2)\|M\|_{\mathcal{A}}$. Why is $T(0)$ set to zero?

D1: Direct

Step 6.6. Bound each term of the source series

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.2, 6.4, 6.5.

Let $X_1 = tI_8$ and $X_{p+1} = t\mathcal{T}X_p$. Prove $\|X_p\|_{\mathcal{A}} \le (3/2)^{p-1}$.

D2: Short proof

Step 6.7. Solve the reduced source equation

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.4, 6.5, 6.6.

Show that $X = \sum_{p\ge1} \rho^p X_p$ converges in the Wiener norm and is the unique Wiener-space solution of $X = H + H\mathcal{T}X$. First sum the norm bounds with $q = 3\rho/2 < 1$, then pass the bounded linear map $M \mapsto H\mathcal{T}M$ through the series. For uniqueness, bound the norm of the difference of two solutions.

D1: Direct

Step 6.8. Bound the fifth-order remainder

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.6, 6.7.

Set $H_* = \langle X\rangle$. Prove $\|H_* - \sum_{p=1}^{5} \rho^p\langle X_p\rangle\| \le (3/2)^5\rho^6/(1-3\rho/2)$.

D1: Direct

Step 6.9. Bound the Green multipliers

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 5.14.

Prove $\|\mathcal{R}(k)\| \le 4/|k|^2$ for $k \ne 0$, and deduce $\|k_\nu k_x \mathcal{R}(k)\| \le 4$ for $\nu = x, y$.

D1: Direct

Step 6.10. Recover the Fourier potential

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.7, 6.9.

For horizontal ζ, solve $(\partial_{xx} + D\partial_{yy})u = \partial_x(PXP^\top\zeta_x)$ in Fourier coefficients with $\widehat{u}(0) = 0$.

D2: Short proof

Step 6.11. Check that the potential is real

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 4.12, 6.6, 6.7, 6.10.

Show that the coefficients in step 6.10 satisfy $\widehat{u}(-k) = \overline{\widehat{u}(k)}$.

D3: Guided technical

Step 6.12. Check admissibility of the potential

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.3, 6.7, 6.9, 6.10, 6.11.

Use steps 6.9–6.10 and $|k| \ge 1$ on $\mathbb{Z}^2 \setminus \{0\}$ to show $u \in H^1_{\mathrm{per}}(Y;\mathbb{R}^{16})$. Bound the absolute sums for u and its formal first derivatives. For the weak-derivative identity, integrate by parts for finite Fourier sums and pass to their uniform limits against a smooth periodic test.

D1: Direct

Step 6.13. Find the reduced horizontal derivative

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.5, 6.10, 6.12.

Multiply the x-derivative formula in step 6.10 by $P^\top$ and prove $P^\top \partial_x u = (\mathcal{T}X)P^\top \zeta_x$.

D3: Guided technical

Step 6.14. Verify the cell equilibrium equation

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 1.5, 4.2, 6.1, 6.7, 6.10, 6.12, 6.13.

Use $X = H + H\mathcal{T}X$ and step 6.13 to prove $\operatorname{div}[\overline{L}(\zeta + \nabla u)] = 0$. Deduce that u is the corrector for $\overline{L}$. Write the two flux rows explicitly and use the baseline equation of step 6.10 in distributions. Extend the test identity from smooth tests to H^1 tests, and use Step 1.5 to identify the energy minimizer.

D2: Short proof

Step 6.15. Average the flux

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 1.1, 1.7, 1.9, 2.8, 2.9, 6.8, 6.14.

For $e = \zeta + \nabla u$, show $\langle \overline{L}e \rangle_x = \zeta_x - PH_*P^\top \zeta_x$ and $\langle \overline{L}e \rangle_y = 0$. Deduce $B = (\overline{L})_* = L(H_*)$. Use Step 1.1 for periodic derivatives and Step 2.8–Step 2.9 to include vertical mean gradients.

D1: Direct

Step 6.16. Prove symmetry of the averaged source

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 1.9, 3.14, 6.15.

Use self-adjointness of $B = L(H_*)$ and $S = P^\top P \succ 0$ to show $H_*^\top = H_*$.

D1: Direct

Step 6.17. Check the first two averaged coefficients

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.2, 6.5, 6.6, 6.7, 6.8.

Use $X_2 = t\mathcal{T}(tI_8)$ to show $H_* = (\rho/2)I_8 + (\rho^2/72)[T(a) + T(c)] + O(\rho^3)$. Here $O(\rho^3)$ means that the operator norm of the error is at most $C\rho^3$ for all sufficiently small positive ρ.

D2: Short proof

Step 6.18. Track the Fourier support

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.2, 6.6.

Prove that every frequency in X_p satisfies $|k_x| \le p$ and $|k_y| \le 3p$.

D3: Guided technical

Step 6.19. Differentiate the source series once

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.3, 6.6, 6.7, 6.18.

Use step 6.18 to prove $\|\partial_x X\|_{\mathcal{A}} \le \rho/(1-q)^2$ and $\|\partial_y X\|_{\mathcal{A}} \le 3\rho/(1-q)^2$, where $q = 3\rho/2$. Use $\sum_{p\ge 1} pq^{p-1} = (1-q)^{-2}$. To justify differentiation, integrate the uniformly convergent derivative series along one coordinate segment and compare with the original series at the segment endpoints.

D1: Direct

Step 6.20. Bound the source field itself

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 3.14, 6.3, 6.7, 6.9, 6.10.

Set $\sigma_\zeta = PP^\top(\zeta_x + \partial_x u)$ and $q = 3\rho/2$. Prove $\|\sigma_\zeta\|_\infty \le \frac{3}{2}[1 + 6\rho/(1-q)]|\zeta_x|$.

D2: Short proof

Step 6.21. Bound the source field derivatives

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.9, 6.10, 6.19, 6.20.

Use step 6.19 and the bounded corrector multipliers to bound $\partial_x \sigma_\zeta$ and $\partial_y \sigma_\zeta$ by constants times $|\zeta_x|$. Explain why the constants remain valid for every stripe count N in the approximation of Module 8.

D3: Guided technical

Step 6.22. Prove smoothness of the fields

Lookup: module 6 guide; setups 6.1, 6.2. *Uses:* 6.3, 6.9, 6.10, 6.12, 6.18, 6.19.

Show that differentiating X_p any fixed number of times adds only a polynomial factor in p. Deduce that X, u, and $e = \zeta + \nabla u$ are smooth. For nonnegative integers r, s, establish $\|\partial_x^r \partial_y^s X_p\|_{\mathcal{A}} \le 3^s p^{r+s} \|X_p\|_{\mathcal{A}}$. For any derivative of u of positive order, leave one x- or y-derivative in the bounded multiplier from step 6.9 and transfer the remaining derivatives to X.

7 The first nonzero energy difference

Prerequisites. Generator identities: Step 3.5–Step 3.6; the energy detector: Step 4.7; the convergent expansion and mean flux: Step 6.6–Step 6.8, Step 6.15.

Definition lookup. The next setups recall all matrices, frequencies, and the expansion. The lookup paragraph defines the Fourier convention, coefficient extraction, permutation sign, and determinant; the identity before Step 7.19 is supplied.

Supplied background. B3, B5, B7. Each key links to its statement, scope, and reference.

Setup 7.1. Work on $Y = (0, 2\pi)^2$ with normalized average $\langle\cdot\rangle$. The real symmetric matrices $\Gamma_1, \dots, \Gamma_4$ satisfy $\Gamma_i^2 = I_8$, $\Gamma_i\Gamma_j = -\Gamma_j\Gamma_i$ for $i \neq j$, and $\Gamma_1\Gamma_2\Gamma_3\Gamma_4 = \Omega = \operatorname{diag}(I_4, -I_4)$. Each Γ_j has zero 4×4 diagonal blocks. Put $\mathcal{S} = \operatorname{span}\{I_8, \Gamma_1, \dots, \Gamma_4\}$ and $d(M) = M_{11} - M_{55}$. Earlier modules identify the measured energy difference as $\ell(B) = d(H_*)$.

Setup 7.2. For $k \neq 0$, use

$$T(k) = \frac{1}{4}\sum_{j=1}^{4} \frac{jk_x^2}{jk_x^2 + k_y^2}(I_8 + \Gamma_j), \qquad T(0) = 0.$$

Thus $T(k) \in \mathcal{S}$, $T(-k) = T(k)$, $T(k) = 0$ when $k_x = 0$, and $\|T(k)\| \le 3/2$. Let $a = (1, 0)$, $b = (0, 1)$, $c = (1, 3)$, and $t = \frac{1}{2} + \frac{1}{6}[\cos x + \cos y + \cos(x + 3y)]$. Its Fourier coefficients are $\widehat{t}(0) = 1/2$, $\widehat{t}(\pm a) = \widehat{t}(\pm b) = \widehat{t}(\pm c) = 1/12$, and zero otherwise. The convergent expansion is

$$H_* = \sum_{p \ge 1} \rho^p \langle X_p \rangle, \qquad X_1 = tI_8, \qquad X_{p+1} = t\mathcal{T}X_p, \quad \widehat{\mathcal{T}M}(k) = T(k)\widehat{M}(k).$$

For $0 < \rho < 2/3$, its remainder $R_6 = H_* - \sum_{p=1}^{5} \rho^p \langle X_p \rangle$ satisfies $\|R_6\| \le (3/2)^5 \rho^6/(1 - 3\rho/2)$.

Setup 7.3. Lookup: $\langle M \rangle = (2\pi)^{-2} \int_Y M$ and $\widehat{M}(k) = \langle M(x, y)e^{-\mathrm{i}k\cdot(x,y)} \rangle$ for $k \in \mathbb{Z}^2$; hence $\widehat{M}(0) = \langle M \rangle$. Matrix norms are operator norms, $\|M\| = \sup_{|v|=1} |Mv|$, and e_i is a coordinate unit vector. A skew-symmetric real matrix satisfies $A^\top = -A$. The sign $\operatorname{sgn}(\pi)$ is $(-1)^m$ if a permutation is put in order by m adjacent swaps, and $\det A = \sum_\pi \operatorname{sgn}(\pi) \prod_i A_{i,\pi(i)}$. Interchanging two rows changes its sign; equal rows make it zero. The notation $[\rho^p]f$ denotes the coefficient of ρ^p in the given convergent power series. A multiset records its elements with multiplicity while ignoring order. A closed path is an ordered list of frequency steps with sum zero; its intermediate vertices are the partial sums K_j. Products keep their written order. All frequency sums range over $\mathbb{Z}^2$. At each fixed order the Fourier sums are finite: X_1 has finite support, a multiplier preserves support, and multiplication by the finite Fourier series of t preserves finiteness.

D1: Direct

Step 7.1. A norm bound

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* local setup only.

Show $|d(M)| \le 2\|M\|$ for every 8×8 matrix M. Use the unit coordinate vectors e_1, e_5.

D1: Direct

Step 7.2. The basic values of the detector

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* local setup only.

Compute $d(I_8)$, $d(\Gamma_j)$, and $d(\Omega)$ from the displayed block information.

D2: Short proof

Step 7.3. Two short skew-symmetric products

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* local setup only.

For distinct indices, prove that $\Gamma_i\Gamma_j$ and $\Gamma_i\Gamma_j\Gamma_k$ are skew-symmetric. Explain why their diagonal entries are zero.

D2: Short proof

Step 7.4. Cancellation for at most three factors

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.2, 7.3.

Let $M_1, \dots, M_q \in \mathcal{S}$ with $q \in \{0, 1, 2, 3\}$. Expand the product and use $\Gamma_j^2 = I_8$ to show $d(M_1 \cdots M_q) = 0$; the empty product is I_8.

D2: Short proof

Step 7.5. A permuted four-generator word

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.2.

For a permutation π of $\{1, 2, 3, 4\}$, show $d(\Gamma_{\pi(1)}\Gamma_{\pi(2)}\Gamma_{\pi(3)}\Gamma_{\pi(4)}) = 2\,\mathrm{sgn}(\pi)$. Count the sign changes when adjacent generators are exchanged.

D2: Short proof

Step 7.6. Four factors produce a determinant

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.4, 7.5.

Write $M_i = \alpha_{i0} I_8 + \sum_{j=1}^{4} \alpha_{ij}\Gamma_j$. Use Exercises 7.4–7.5 to show

$$d(M_1 M_2 M_3 M_4) = 2 \det[\alpha_{ij}]_{i,j=1}^{4}.$$

Hint: only words containing all four different generators survive.

D2: Short proof

Step 7.7. Why fourth order is still zero

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 6.6, 7.4.

A term in $\langle X_p \rangle$ contains $p-1$ multiplier matrices T. Use Exercise 7.4 to show that the coefficients of $\rho, \rho^2, \rho^3, \rho^4$ in $\ell(B)$ vanish.

D1: Direct

Step 7.8. One convolution step

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* local setup only.

Use $X_{p+1} = t\mathcal{T}X_p$ and the finite-sum Fourier product rule $\widehat{tM}(q) = \sum_s \widehat{t}(s)\widehat{M}(q-s)$ to show

$$\widehat{X}_{p+1}(q) = \sum_{s} \widehat{t}(s) T(q-s) \widehat{X}_p(q-s)$$

while keeping each multiplier on the left of $\widehat{X}_p$.

D3: Guided technical

Step 7.9. The ordered fifth coefficient

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.8.

Iterate Exercise 7.8 at $q = 0$, labeling the outermost frequency k_1. With $K_j = k_1 + \cdots + k_j$ and $k_5 = -K_4$, verify

$$\widehat{X}_5(0) = \sum_{k_1, \dots, k_4} \left(\prod_{i=1}^{5} \widehat{t}(k_i) \right) T(-K_1) T(-K_2) T(-K_3) T(-K_4).$$

Start by writing $\widehat{X}_2(0)$; each further substitution adds one factor on the right.

D1: Direct

Step 7.10. Rewrite it as a closed path

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.9.

Use $T(-k) = T(k)$, $\langle X_5 \rangle = \widehat{X}_5(0)$, and $\ell(B) = d(H_*)$ in Exercise 7.9 to obtain

$$[\rho^5]\ell(B) = \sum_{k_1+\cdots+k_5=0} \left(\prod_{i=1}^{5} \widehat{t}(k_i) \right) d(T(K_1)T(K_2)T(K_3)T(K_4)).$$

Explain why the five steps have total frequency zero.

D1: Direct

Step 7.11. Zero endpoint steps

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.10.

Show that a summand in Exercise 7.10 vanishes if $k_1 = 0$ or $k_5 = 0$.

D2: Short proof

Step 7.12. Zero interior steps

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.6, 7.10.

Show that a summand vanishes if one of k_2, k_3, k_4 is zero. Use the determinant rule and the repeated partial sums.

D2: Short proof

Step 7.13. The signed-count equations

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.10, 7.11, 7.12.

For five steps in $\{\pm a, \pm b, \pm c\}$, let n_a be the number of a steps minus the number of $-a$ steps, and define n_b, n_c similarly. Show that a closed path satisfies $n_a = -n_c$, $n_b = -3n_c$, and $5 \geq 5|n_c|$. For the inequality, compare the total number of steps in each direction with the magnitude of its signed count.

D2: Short proof

Step 7.14. Only two multisets remain

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.13.

Show that $n_c = 0$ is impossible by parity. Deduce that the multiset of steps is $\{a, b, b, b, -c\}$ or its negative.

D2: Short proof

Step 7.15. Only four ordered paths remain

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.10, 7.14.

Use $T(\pm b) = 0$ to show that the only possibly nonzero paths are $(a, b, b, b, -c)$, $(-c, b, b, b, a)$ and their negatives.

D1: Direct

Step 7.16. The four intermediate directions

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.15.

For $(a, b, b, b, -c)$, compute K_1, K_2, K_3, K_4 and their squared slopes $(K_{j,y}/K_{j,x})^2$.

D2: Short proof

Step 7.17. The signs agree

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.6, 7.15, 7.16.

Show that exchanging the two endpoint steps reverses the four squared slopes, and that negating all steps leaves T unchanged. Conclude that the four surviving contributions have the same sign and magnitude.

D2: Short proof

Step 7.18. Count the numerical factors

Lookup: module 7 guide; setups 7.1, 7.2, 7.3. *Uses:* 7.6, 7.10, 7.11, 7.12, 7.15, 7.16, 7.17.

Let $r = (0, 1, 4, 9)$. Account separately for four paths, five Fourier coefficients, and the determinant rule to prove

$$[\rho^5]\ell(B) = \frac{8}{12^5\,4^4}\det\left[\frac{j}{j+r_i}\right]_{i,j=1}^{4}.$$

Setup 7.4. For the next two exercises you may use the Cauchy determinant identity

$$\det\left[\frac{1}{x_i+y_j}\right]_{i,j=1}^{4} = \frac{\prod_{i<k}(x_k-x_i)\prod_{j<l}(y_l-y_j)}{\prod_{i,j}(x_i+y_j)}.$$

The parameters used below are distinct and all denominators are positive (B7).

D1: Direct

Step 7.19. Determine the determinant's sign

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 7.16.

Apply the supplied identity with $x_i = r_i$ and $y_j = j$. Show that $\det[j/(j+r_i)] > 0$ without expanding a 4×4 determinant.

D1: Direct

Step 7.20. Evaluate the constant

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 7.18, 7.19.

Use $\prod_{i<k}(r_k - r_i) = 4320$ and $\prod_{j<l}(l-j) = 12$ to verify

$$\det\left[\frac{j}{j+r_i}\right] = \frac{3}{200200}, \qquad \kappa := \frac{4}{12^5\,4^4}\,\frac{3}{200200} = \frac{1}{1062744883200}.$$

D1: Direct

Step 7.21. Bound the energy remainder

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 6.8, 7.1, 7.7, 7.18, 7.20.

Use Exercise 7.1 and the supplied bound on R_6 to show

$$\ell(B) \geq 2\kappa\rho^5 - \frac{2(3/2)^5\rho^6}{1-3\rho/2}.$$

D1: Direct

Step 7.22. Fix a contrast with a strict difference

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 7.20, 7.21.

Set $\rho = \kappa/32$. Use $\rho < 1/3$ in Exercise 7.21 to prove $\ell(B) \geq (269/256)\kappa\rho^5 > 0$.

D1: Direct

Step 7.23. Why the smooth medium is not binary

Lookup: module 7 guide; setups 7.1, 7.2, 7.3, 7.4. *Uses:* 4.4, 7.22.

The smooth coefficient is $\overline{L} = (1-t)L_0 + tL_1$. Explain in one or two sentences why its positive energy difference does not yet provide a composite using only the two coefficient values L_0, L_1.

8 Replace the smooth medium by binary stripes

Prerequisites. The corrector and mean flux: Step 1.4, Step 1.6, Step 1.9, Step 1.12; smooth-field bounds: Step 6.20–Step 6.22; strict positivity at fixed contrast: Step 7.22.

Definition lookup. The setups below define the stripe geometry, corrector fields, and residual constants. The lookup recalls the loads and spatial-row notation, and supplies the weak-derivative and Lipschitz facts needed at the interfaces.

Supplied background. B1, B6, B8, B11. Each key links to its statement, scope, and reference.

Setup 8.1. On $Y = (0, 2\pi)^2$, use $t = \frac{1}{2} + \frac{1}{6}[\cos x + \cos y + \cos(x+3y)]$ and fix the positive contrast from Exercise 7.22. The smooth coefficient is $\overline{L} = \operatorname{diag}(I_{16} - \rho t P P^\top, D)$, its effective tensor is B, and $\Delta = \ell(B) > 0$, where $\ell(C) = \eta \cdot C\eta - \xi \cdot C\xi$. The two phases are $L_0 = \operatorname{diag}(I_{16}, D)$ and $L_1 = \operatorname{diag}(I_{16} - \rho P P^\top, D)$. Their bounds are $\frac{1}{4} I \preceq L_i \preceq I$. For either selected mean gradient $\zeta \in \{\xi, \eta\}$, let $e = \zeta + \nabla u_\zeta$ be the smooth corrector gradient, $j = \overline{L} e$, and $\sigma = P P^\top e_x$. Thus $\operatorname{div} j = 0$ and $\langle j \rangle = B\zeta$. The preceding regularity argument supplies finite constants $S_0 = \|\sigma\|_\infty$, $S_x = \|\partial_x \sigma\|_\infty$, $S_y = \|\partial_y \sigma\|_\infty$, depending on ζ but independent of the stripe count. The fields e, j, σ are periodic.

Setup 8.2. For an integer $N \geq 2$, put $h = 2\pi/N$ and $y_k = kh$ for $0 \leq k \leq N$. For $k = 0, \dots, N-1$, set $t_k(x) = t(x, y_k)$. In $y_k \leq y < y_{k+1}$, set $\tau = (y - y_k)/h$, $\chi_N = \mathbf{1}_{\{\tau < t_k(x)\}}$, and $L_N = (1 - \chi_N) L_0 + \chi_N L_1$; extend periodically. Let B_N denote its effective tensor. All L^2 norms below are normalized: $\|f\|_2^2 = \langle |f|^2 \rangle$.

Setup 8.3. Lookup: $P \in \mathbb{R}^{16 \times 8}$ is the stacked matrix of Step 3.11, $S = P^\top P$, and $D = \operatorname{diag}(1, 1/2, 1/3, 1/4) \otimes I_4$, where $\otimes I_4$ repeats each listed scalar as a 4×4 block. The loads from Step 4.7 have $\xi_x = P S^{-1} e_1$, $\eta_x = P S^{-1} e_5$, and $\xi_y = \eta_y = 0$. A field $E \in \mathbb{R}^{2 \times 16}$ has transposed row vectors $E_x, E_y \in \mathbb{R}^{16}$, norm $|E|^2 = |E_x|^2 + |E_y|^2$, and product $E \cdot J = E_x \cdot J_x + E_y \cdot J_y$. The block tensor $\operatorname{diag}(A, D)$ maps these rows to AE_x, DE_y. The order $A \preceq B$ means $v \cdot Av \leq v \cdot Bv$ for every v, and coefficient norms are operator norms. The indicator $\mathbf{1}_A$ is 1 on A and 0 elsewhere; $0 \leq t \leq 1$ follows from $|\cos| \leq 1$. The notation a.e. means outside a set of area zero, and $\|f\|_\infty$ is its essential supremum.

Setup 8.4. Weak-derivative lookup and supplied background: $g = \partial_\nu f$ means $\langle g \cdot \varphi \rangle = -\langle f \cdot \partial_\nu \varphi \rangle$ for every smooth periodic test φ. The distributional derivative of f is the linear functional $\varphi \mapsto -\langle f \cdot \partial_\nu \varphi \rangle$, even when this functional is not represented by an ordinary function. Applying a second derivative moves both derivatives onto the test with positive sign. Periodic H^1 consists of L^2 functions whose weak first derivatives are in L^2, with matching opposite traces; a trace is the boundary value in this Sobolev sense. Smooth periodic functions are dense in this space (B1, B6). Supplied Lipschitz facts: a Lipschitz function has a.e. first derivatives that are its weak derivatives; the minimum of two Lipschitz functions is Lipschitz; the usual product rule holds with a smooth factor (B8). Stripwise integration by parts is valid for these functions. For continuous g, the graph $y = g(x)$ is measurable and has area zero: each vertical section has length zero, so Fubini's theorem applies (B11). These background facts need not be reproved.

D1: Direct

Step 8.1. Count the phase inside one strip

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* local setup only.

For each fixed x, show that the integral of $\chi_N(x, \cdot)$ across strip k is $h t_k(x)$. Conclude that L_N takes only the original two phase values.

D2: Short proof

Step 8.2. Count the total fraction exactly

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.1.

Integrate Exercise 8.1 in x and use $\sum_{k=0}^{N-1} \cos(2\pi k/N) = 0$ to prove $\langle \chi_N \rangle = 1/2$.

D1: Direct

Step 8.3. Integrate the zero-mean oscillation

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.1.

Define $w_N(x,y) = \int_{y_k}^{y} (\chi_N(x,s) - t_k(x))\,\mathrm{d}s$ in strip k. Show that $w_N = h[\min\{\tau, t_k\} - \tau t_k]$ and that it vanishes at both strip edges.

D1: Direct

Step 8.4. Bound the primitive

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.3.

For $0 \le s, \tau \le 1$, show $0 \le \min\{\tau, s\} - \tau s \le s(1-s) \le 1/4$. Deduce $|w_N| \le h/4$.

D2: Short proof

Step 8.5. Differentiate inside a strip

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.3.

Differentiate the two formulas on $\tau < t_k(x)$ and $\tau > t_k(x)$ to prove a.e. that $\partial_y w_N = \chi_N - t_k$ and $|\partial_x w_N| \le h\|t_x\|_\infty$. The omitted equality set is the graph $y = y_k + h t_k(x)$ and has area zero. Use the supplied Lipschitz facts to identify the derivatives as weak derivatives inside a strip.

D1: Direct

Step 8.6. Bound the sampling error

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* local setup only.

Use the mean value theorem in the y coordinate to show $|t_k(x) - t(x,y)| \le h\|t_y\|_\infty$ throughout strip k.

D3: Guided technical

Step 8.7. Check the derivatives across strips

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.3, 8.4, 8.5.

Explain why the matching zero traces of w_N at strip edges make the derivatives in Exercise 8.5 its weak derivatives on the whole periodic cell. Test against a smooth periodic φ: integrate $w_N \partial_y \varphi$ strip by strip and check every horizontal boundary term, including the top and bottom cell edges. For ∂_x, integrate each strip in x and use periodicity. The bounded piecewise derivatives then show $w_N \in H^1_{\mathrm{per}}$.

D1: Direct

Step 8.8. Compute the flux mismatch

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* local setup only.

At the unchanged gradient e, show $(j - L_N e)_x = \rho(\chi_N - t)\sigma$ and $(j - L_N e)_y = 0$.

D3: Guided technical

Step 8.9. Preserve equilibrium

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 6.22, 8.7.

Define the flux with transposed row vectors

$$j_N = j + \rho \begin{pmatrix} [-\partial_y(w_N\sigma)]^\top \\ [\partial_x(w_N\sigma)]^\top \end{pmatrix}.$$

Prove $\operatorname{div} j_N = 0$ in distributions by cancellation of mixed derivatives. Test the correction against a smooth periodic vector function, move each derivative onto the test, and use $\partial_{xy}\varphi = \partial_{yx}\varphi$.

D2: Short proof

Step 8.10. Preserve the mean flux

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.9.

Show $\langle j_N \rangle = B\zeta$ by averaging the periodic derivatives in Exercise 8.9.

D1: Direct

Step 8.11. Cancel the horizontal oscillation

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.5, 8.8, 8.9.

For $r_N = j_N - L_N e$, use $\partial_y w_N = \chi_N - t_k$ to derive the a.e. identity

$$(r_N)_x = \rho[(t_k - t)\sigma - w_N\partial_y\sigma].$$

D1: Direct

Step 8.12. Compute the vertical residual

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.8, 8.9, 8.11.

Use the product rule to show a.e. that $(r_N)_y = \rho[(\partial_x w_N)\sigma + w_N\partial_x\sigma]$.

D1: Direct

Step 8.13. Make the residual small

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.4, 8.5, 8.6, 8.11, 8.12.

Put $C_\zeta = (\|t_x\|_\infty + \|t_y\|_\infty)S_0 + (S_x + S_y)/4$. Use Exercises 8.4–8.6 to prove $\|r_N\|_2 \le \rho h C_\zeta$.

D3: Guided technical

Step 8.14. Obtain the error orthogonality

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 1.4, 1.6, 8.9, 8.11.

Let $e_N = \zeta + \nabla u_{N,\zeta}$ be the exact cell gradient for L_N, and put $\delta e = e_N - e$. Show $\langle \delta e \cdot (L_N \delta e - r_N) \rangle = 0$ using equilibrium of $L_N e_N$ and j_N. Their difference is an L^2 flux; first test it with smooth periodic potentials, then approximate $u_{N,\zeta} - u_\zeta$ in H^1 and pass to the limit by Cauchy–Schwarz.

D1: Direct

Step 8.15. Control the gradient error

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.14.

Use Exercise 8.14, $L_N \succeq \frac{1}{4} I$, and Cauchy–Schwarz to prove $\|e_N - e\|_2 \le 4\|r_N\|_2$. Include the case $\delta e = 0$.

D2: Short proof

Step 8.16. Control the selected mean flux

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 1.9, 8.10, 8.13, 8.15.

Average $L_N e_N - j_N = L_N \delta e - r_N$ and use $\|L_N(x,y)\| \le 1$ a.e. to show $|(B_N - B)\zeta| \le 5\|r_N\|_2 \le 10\pi\rho C_\zeta / N$.

D1: Direct

Step 8.17. Control the measured energy difference

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 8.16.

Use Exercise 8.16 at ξ and η to prove $|\ell(B_N) - \Delta| \le K_\rho/N$, where $K_\rho = 10\pi\rho(|\xi|C_\xi + |\eta|C_\eta)$.

D1: Direct

Step 8.18. Choose one finite stripe count

Lookup: module 8 guide; setups 8.1, 8.2, 8.3, 8.4. *Uses:* 4.7, 5.16, 5.17, 7.22, 8.2, 8.17.

Choose an integer $N \ge 2$ with $N > 2K_\rho/\Delta$ and prove $\ell(B_N) > \Delta/2$. Explain why the contrast is held fixed while making this choice.

9 Prescribed fractions and more potentials

Prerequisites. Binary witness: Step 8.18; arithmetic horizontal layers: Step 2.10; detector continuity and laminate limits: Step 4.8, Step 5.17.

Definition lookup. The setup defines C_θ's ingredients and the supplied recovery inputs. The lookup below defines cell gradients, exact-fraction closure, and direct sums; Step 9.1–Step 9.2 define C_θ.

Supplied background. B4, B9, B11, B12. Each key links to its statement, scope, and reference.

Setup 9.1. Manuscript: Section 5.3. Let $0 < \theta < 1$ be the prescribed phase-one fraction. The preceding modules give constituents L_0, L_1 and a periodic binary composite with tensor B_N, phase-one fraction $1/2$, and $\ell(B_N) = g_N > 0$. All three tensors have the same constant vertical block D, so horizontal layering gives their arithmetic mean. The detector satisfies $\ell(L_i) = 0$ and $|\ell(A) - \ell(B)| \le (8/3)\|A - B\|$.

Use the following background results. **Separated-scale periodic homogenization (B4):** layering periodic composites and inserting increasingly fine copies of their original phase patterns produces periodic binary coefficients whose effective tensors converge to the laminate tensor and whose fractions converge to the weighted fraction. Iteration gives recovery for finite hierarchies; rational-direction approximation and continuity of the laminate formula allow arbitrary layer normals. **Higher integrability (B9):** for periodic self-adjoint coefficients on $\mathbb{R}^{2\times 16}$ with $\mu I \preceq A \preceq MI$, $0 < \mu \le M$, there are $\delta > 0$ and C depending only on these bounds and the dimensions such that $\|F + \nabla u_{A,F}\|_{2+\delta} \le C|F|$. The steps below use these results to correct the fraction exactly.

Setup 9.2. Lookup: $Y = (0, 2\pi)^2$. In Step 9.4–Step 9.6, A, B are measurable periodic self-adjoint coefficient fields on $\mathbb{R}^{2\times 16}$ with common bounds $\mu I \preceq A, B \preceq MI$, $0 < \mu \le M$. Write $e_{A,F} = F + \nabla u_{A,F}$ for the minimizing cell gradient (Step 1.4–Step 1.5) and A_* for its effective tensor, so $F \cdot A_* F = \langle e_{A,F} \cdot A e_{A,F} \rangle$ (Step 1.10). The indicator $\mathbf{1}_E$ equals one on E and zero elsewhere; $\|f\|_p = (\langle |f|^p \rangle)^{1/p}$. The set G_θ is the operator-norm closure of periodic binary effective tensors at exact fraction θ (Step 2.11–Step 2.12). A direct sum $C \oplus I$ acts by C on the original sixteen potential columns and by the identity on the extra columns. For positive-definite C_1, C_2, a unit normal n, and $0 \le \lambda \le 1$, $\mathrm{Lam}_n(C_1, C_2; \lambda)$ denotes the laminate tensor of Step 2.7.

D1: Direct

Step 9.1. Lower the fraction

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 2.10, 8.18.

For $0 < \theta \le 1/2$, mix a fraction λ of B_N with phase 0. Find λ and the resulting tensor C_θ.

D1: Direct

Step 9.2. Raise the fraction

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 2.10, 8.18.

For $1/2 < \theta < 1$, mix a fraction λ of B_N with phase 1. Find λ and C_θ.

D1: Direct

Step 9.3. Keep a positive difference

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.1, 9.2, 4.8.

Use linearity of ℓ to show $\ell(C_\theta) = 2\min(\theta, 1 - \theta) g_N > 0$ for $0 < \theta < 1$.

D2: Short proof

Step 9.4. Compare two cell energies

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 1.5, 1.10.

Let the periodic coefficients A, B obey the common bounds above and agree outside a measurable set $E \subset Y$. Using the B-corrector as a trial for A, prove $F \cdot (A_* - B_*)F \le 2M\langle \mathbf{1}_E | e_{B,F}|^2\rangle$.

D2: Short proof

Step 9.5. Estimate an integral over a small set

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.4.

Put $\alpha = \delta/(2+\delta)$. Use Hölder's inequality and the given higher-integrability bound to show $\langle \mathbf{1}_E | e_{B,F}|^2\rangle \le C^2|F|^2(|E|/|Y|)^\alpha$.

D2: Short proof

Step 9.6. Obtain a uniform tensor estimate

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.4, 9.5, 1.9.

Exchange A and B in the last two steps. Deduce $\|A_* - B_*\| \le 2MC^2(|E|/|Y|)^\alpha$.

D2: Short proof

Step 9.7. Correct a binary fraction exactly

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 2.11.

A binary indicator has fraction θ_n. Explain how to change it on a set of normalized measure $|\theta_n - \theta|$ so the new fraction is exactly θ, even when θ is irrational. Hint: slice the phase to be removed by $\{0 < x < a\}$ and use continuity of its measure in a (B11, B12).

D2: Short proof

Step 9.8. Pass to exact-fraction recovery

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.6, 9.7.

Separated-scale recovery supplies tensors $A_{n,*} \to C_\theta$ and fractions $\theta_n \to \theta$. Use the last two steps to show $C_\theta \in G_\theta(L_0, L_1)$.

D1: Direct

Step 9.9. Select a periodic witness

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.8, 9.3, 4.8.

Let $g_\theta = \ell(C_\theta) > 0$. Show any exact-fraction periodic approximant B_χ with $\|B_\chi - C_\theta\| < 3g_\theta/16$ satisfies $\ell(B_\chi) > g_\theta/2$.

D1: Direct

Step 9.10. Exclude every laminate limit

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 9.9, 5.17.

All finite laminate tensors satisfy $\ell = 0$. Explain why the tensor B_χ just chosen cannot be a limit of them, even if their fractions vary.

D2: Short proof

Step 9.11. Add uncoupled potentials in the cell problem

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 1.1, 1.5.

For $m > 16$ put $L_i^{(m)} = L_i \oplus I$ on the original and extra columns. Show the effective tensor of the same indicator is $B_\chi \oplus I$.

D2: Short proof

Step 9.12. Add uncoupled potentials in a laminate

Lookup: module 9 guide; setups 9.1, 9.2. *Uses:* 2.2, 2.7, 9.11.

Show a laminate of $C_1 \oplus I, C_2 \oplus I$ is $\mathrm{Lam}_n(C_1, C_2; \lambda) \oplus I$ for every normal n.

10 Complementary energy and planar duality

Prerequisites. The ordered phases and detecting loads: Step 4.4–Step 4.7; layer compatibility/equilibrium: Step 2.1–Step 2.7; Fourier weak derivatives: Step 6.10–Step 6.12.

Definition lookup. The setup defines rotation, curl, the planar dual, and the two quadratic branches. The lookup below recalls the exact phase matrices and explains Fourier and operator notation.

Supplied background. B3, B4, B6, B10. Each key links to its statement, scope, and reference.

Setup 10.1. Manuscript: Section 6.1 and Proposition 6.2. Tensors act on $E = \mathbb{R}^{2\times 16}$. For a self-adjoint positive-definite tensor C, define its planar dual by

$$R_\perp = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C^{\mathrm{d}} = R_\perp C^{-1} R_\perp^\top.$$

Rotation acts on spatial rows and satisfies $R_\perp^\top R_\perp = I$. The fields j, w below belong to $L^2_{\mathrm{per}}(Y;E)$. Define curl columnwise by $\operatorname{curl} w = \partial_x w_y - \partial_y w_x$, with derivatives understood in distributions. The Fourier convention is $\widehat{w}(k) = \langle w(x) e^{-\mathrm{i}k\cdot x}\rangle$. Parseval and Fourier differentiation are supplied in B6.

For $F = (f, g) \in \mathbb{R}^{2\times 32}$, define $q_i(F) = f \cdot L_i f + g \cdot L_i^{\mathrm{d}} g$ and $W = \min(q_0, q_1)$; the same index $i \in \{0, 1\}$ chooses both terms. A laminate has a unit normal n and proportions $\lambda, 1-\lambda$, $0 \le \lambda \le 1$. Its primal gradients have continuous tangential rows and its fluxes have continuous normal rows. Denote its effective tensor by C_λ.

Setup 10.2. Lookup: $Y = (0, 2\pi)^2$, $k \in \mathbb{Z}^2$, and the dot product $k \cdot \widehat{w}(k)$ contracts spatial rows separately in every potential column. Curl and divergence are distributional derivatives as in Module 8's lookup (B8). The statement $\nabla v = w$ will be verified by finite Fourier truncations and an H^1 limit; $\widehat{v}(0) = 0$ fixes the mean. An expression such as $R_\perp C^{-1} R_\perp^\top$ denotes composition of operators on the spatial-first gradient space, in the written order. From Step 3.11–Step 4.7, $S = P^\top P$, P has full column rank, $L_0 = \operatorname{diag}(I_{16}, D)$, $L_1 = \operatorname{diag}(I_{16} - \rho P P^\top, D)$, $0 < \rho < 2/3$, $D = \operatorname{diag}(1, 1/2, 1/3, 1/4) \otimes I_4$, and $\xi_x = P S^{-1} e_1$, $\xi_y = 0$. Thus $P^\top \xi_x = e_1 =: e_+$.

D1: Direct

Step 10.1. See why the original minimum is trivial

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 4.4.

Given $L_1 \preceq L_0$, prove $\min(f \cdot L_0 f, f \cdot L_1 f) = f \cdot L_1 f$.

D1: Direct

Step 10.2. Express complementary energy as a gradient energy

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 5.14.

For positive-definite C, show C^{d} is positive definite and $(R_\perp p) \cdot C^{\mathrm{d}}(R_\perp p) = p \cdot C^{-1} p$.

D1: Direct

Step 10.3. Rotate equilibrium into zero curl

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 1.6, 1.12.

If $\operatorname{div} j = 0$ and $p = \langle j \rangle$, show $w = R_\perp(j - p)$ has zero curl and zero mean.

D2: Short proof

Step 10.4. Construct the stream potential in Fourier variables

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.3, 6.10.

For a mean-zero, curl-free $w \in L^2_{\mathrm{per}}(Y;E)$, set $\widehat{v}(k) = -\mathrm{i}\, k \cdot \widehat{w}(k)/|k|^2$ for $k \neq 0$ and $\widehat{v}(0) = 0$. Show $\mathrm{i}k\widehat{v}(k) = \widehat{w}(k)$ for all k. Hint: zero curl makes each column of $\widehat{w}(k)$ parallel to k.

D3: Guided technical

Step 10.5. Check regularity and reality of the stream potential

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.4, 6.11, 6.12.

For real $w \in L^2$ in the preceding step, show the constructed v is real and belongs to H^1_{per}. Use Parseval with weight $1 + |k|^2$; finite Fourier truncations justify the weak derivative.

D1: Direct

Step 10.6. Verify the dual constitutive law

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.2.

For a primal layer with $j = Ce$, set $e^{\mathrm{d}} = R_\perp j$ and $j^{\mathrm{d}} = R_\perp e$. Check $j^{\mathrm{d}} = C^{\mathrm{d}} e^{\mathrm{d}}$.

D2: Short proof

Step 10.7. Rotate the interface conditions

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 2.5, 10.3, 10.6.

Put $\tau = R_\perp n$. Show continuity of $n^\top j$ and $\tau^\top e$ implies continuity of $\tau^\top e^{\mathrm{d}}$ and $n^\top j^{\mathrm{d}}$.

D2: Short proof

Step 10.8. Identify the effective tensor of the dual layers

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 1.9, 2.1, 2.4, 2.5, 2.6, 10.6, 10.7.

Primal layer means are $\langle e \rangle = f$ and $\langle j \rangle = C_\lambda f$. Show the effective tensor of the dual layers is C^{d}_λ.

D1: Direct

Step 10.9. Check what the two branches mean

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.2, 4.2.

Show each $q_i(f, g) = f \cdot L_i f + g \cdot L_i^{\mathrm{d}} g$ is a positive-definite homogeneous quadratic with its only minimum at zero.

D1: Direct

Step 10.10. The primal term prefers phase one

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 4.4, 4.5.

Use $L_0 - L_1 = \mathrm{diag}(\rho P P^\top, 0)$ and $P^\top \xi_x = e_+$ to show $q_1(\xi, 0) < q_0(\xi, 0)$.

D2: Short proof

Step 10.11. The dual term prefers phase zero

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 3.12, 4.5, 10.2.

On $\operatorname{range} P$, $PP^\top$ has eigenvalues $1/2, 3/2$. Use $\xi_x \in \operatorname{range} P \setminus \{0\}$ and $A_1 = I - \rho P P^\top$ to show $q_0(0, R_\perp \xi) < q_1(0, R_\perp \xi)$.

D1: Direct

Step 10.12. Keep the phase index shared

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 10.2, 10.9.

For the scalar example $L_0 = 2I_2, L_1 = I_2$ and $|f| = |g| = 1$, compare $\min_i [f \cdot L_i f + g \cdot L_i^{\mathrm{d}} g]$ with $\min_i f \cdot L_i f + \min_i g \cdot L_i^{\mathrm{d}} g$.

D1: Direct

Step 10.13. Keep the coefficients rational

Lookup: module 10 guide; setups 10.1, 10.2. *Uses:* 4.4, 10.2.

If L_i has rational entries and is invertible, explain why q_i has rational coefficients.

11 A rank-one convex lower bound

Prerequisites. Compact family and invariant: Step 4.2–Step 4.8; lamination stability: Step 5.13–Step 5.17; dual lamination: Step 10.6–Step 10.8.

Definition lookup. The setup defines E_C, V, rank-one convexity, and the envelope $\mathrm{RC}\,W$. Rotation and the dual are recalled below.

Supplied background. B3, B10, B12. Each key links to its statement, scope, and reference.

Setup 11.1. Manuscript: Proposition 6.2. Write $F = (f, g) \in \mathbb{R}^{2\times 32}$, with $f, g \in \mathbb{R}^{2\times 16}$. The symbol p denotes a constant matrix in $\mathbb{R}^{2\times 16}$. Modules 4–5 give a nonempty compact family $\mathcal{K}$ containing L_0, L_1, stable under simple lamination, and satisfying $\mu I \preceq C \preceq MI$ with $0 < \mu \le M$. Its energy invariant is $\xi \cdot C\xi = \eta \cdot C\eta$. Define

$$E_C(f, g) = f \cdot Cf + g \cdot C^{\mathrm{d}} g, \qquad V(F) = \min_{C\in\mathcal{K}} E_C(F).$$

Step 11.2 verifies attainment of this minimum. A finite real-valued function h is rank-one convex if

$$h(\lambda F_1 + (1-\lambda)F_2) \le \lambda h(F_1) + (1-\lambda)h(F_2)$$

for $0 \le \lambda \le 1$ whenever $F_1 - F_2 = n \otimes z$, where $|n| = 1$ and $z \in \mathbb{R}^{32}$. These jumps include zero ($z = 0$); a nonzero rank-one factorization is normalized by replacing n, z with $n/|n|, |n|z$. The envelope $\mathrm{RC}\,W$ is the pointwise supremum of finite real-valued rank-one convex minorants of W. Module 10 shows that the laminate of the duals equals the dual of the primal laminate with the same normal and proportions.

Setup 11.2. Lookup: $F = (f, g)$ denotes two 2×16 column blocks, so $|F|^2 = |f|^2 + |g|^2$. Rotation is $R_\perp = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ on spatial rows and $C^{\mathrm{d}} = R_\perp C^{-1} R_\perp^\top$ (Step 10.2). The obstacle is $W(F) = \min_{i=0,1} E_{L_i}(F)$ (Step 10.9). Compactness and the extreme value theorem are the elementary background in B12; continuity of inversion is explicitly proved in Step 11.2. The norm of a tensor is its operator norm, and all order comparisons are between self-adjoint positive-definite tensors.

D1: Direct

Step 11.1. Bound the paired matrices uniformly

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 4.2, 10.2, 5.14.

Put $K_0 = \max(M, \mu^{-1})$. Show the matrix defining E_C has operator norm at most K_0, uniformly over $C \in \mathcal{K}$.

D2: Short proof

Step 11.2. Show the minimum is attained

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 4.3, 11.1.

Show that $V(F)$ is attained by a tensor $C_F \in \mathcal{K}$.

D1: Direct

Step 11.3. Check homogeneity

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.2.

Prove $V(sF) = s^2 V(F)$ for every real s.

D2: Short proof

Step 11.4. Continuity of the minorant

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.1, 11.2.

Show $|V(F) - V(G)| \le K_0(|F| + |G|)|F - G|$. Hint: use a minimizing tensor at one endpoint, then exchange the endpoints.

D1: Direct

Step 11.5. Compare the minorant with the obstacle

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 4.4, 10.9.

Use $L_0, L_1 \in \mathcal{K}$ to prove $V \le W$.

D1: Direct

Step 11.6. Split an arbitrary rank-one jump

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 2.1, 2.2.

If $F_1 - F_2 = n \otimes z$ in $\mathbb{R}^{2\times 32}$ with $|n| = 1$, write $z = (a, b) \in \mathbb{R}^{16} \times \mathbb{R}^{16}$. Identify the jumps and weighted means in the f and g blocks.

D2: Short proof

Step 11.7. Apply the two laminate trial inequalities

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.6, 2.7, 10.8.

Set $C_\lambda = \mathrm{Lam}_n(C_1, C_2; \lambda)$ for $C_1, C_2 \in \mathcal{K}$. Put $F_\lambda = \lambda F_1 + (1-\lambda)F_2$. For the compatible fields in the preceding step, prove $E_{C_\lambda}(F_\lambda) \le \lambda E_{C_1}(F_1) + (1-\lambda)E_{C_2}(F_2)$.

D2: Short proof

Step 11.8. Conclude rank-one convexity

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.2, 11.5, 11.7, 5.13, 5.15.

Choose C_i attaining $V(F_i)$ in the preceding step. Show V is rank-one convex and hence $V \le \mathrm{RC}\, W$.

D2: Short proof

Step 11.9. Complete the square

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 4.7, 10.2.

Use $\xi \cdot C\xi = \eta \cdot C\eta$ to prove

$$E_C(\xi, R_\perp p) = 2\eta \cdot p + (p - C\eta) \cdot C^{-1}(p - C\eta).$$

D1: Direct

Step 11.10. Obtain a lower bound for the rank-one convex envelope

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.8, 11.9.

Deduce $\mathrm{RC}\, W(\xi, R_\perp p) \ge V(\xi, R_\perp p) \ge 2\eta \cdot p$ for every p.

D2: Short proof

Step 11.11. Understand equality for the minorant

Lookup: module 11 guide; setups 11.1, 11.2. *Uses:* 11.2, 11.9, 11.10.

Show $V(\xi, R_\perp p) = 2\eta \cdot p$ if and only if some $C \in \mathcal{K}$ satisfies $p = C\eta$.

12 A periodic trial and the strict envelope gap

Prerequisites. Exact witness: Step 9.9; stream potential: Step 10.3–Step 10.5; lower bound: Step 11.10.

Definition lookup. The setup defines quasiconvex minorants, the test convention, $\mathrm{QC}\,W, p, g_\chi, \overline{F}$, and the two cell fields. The lookup below recalls the obstacle and the meaning of strong convergence.

Supplied background. B1, B6, B10, B11. Each key links to its statement, scope, and reference.

Setup 12.1. Manuscript: Section 6.3. Keep $W = \min(q_0, q_1)$ on $\mathbb{R}^{2\times 32}$. A finite real-valued, Borel measurable, locally bounded function h is quasiconvex if

$$h(F) \le |Q|^{-1} \int_Q h(F + \nabla\psi), \qquad Q = (0,1)^2,$$

for every $F \in \mathbb{R}^{2\times 32}$ and every Lipschitz $\psi : Q \to \mathbb{R}^{32}$ that vanishes on the boundary. The envelope $\mathrm{QC}\,W$ is the pointwise supremum of these functions with $h \le W$. Such functions are locally Lipschitz and rank-one convex, and their Jensen inequality holds for periodic $W^{1,\infty}$ tests (B10). In particular,

$$h(F) \le \langle h(F + \nabla\varphi)\rangle$$

for smooth periodic $\varphi : Y \to \mathbb{R}^{32}$. The test field has values in a compact set, so the integrand is bounded. Periodic smoothing (B1) gives smooth $\varphi_k \to \varphi$ strongly in H^1 for every $\varphi \in H^1_{\mathrm{per}}(Y;\mathbb{R}^{32})$.

Choose the binary witness χ from Module 9, with effective tensor B_χ and $g_\chi = \ell(B_\chi) > 0$. For $\zeta \in \{\xi, \eta\}$ let $e_\zeta = \zeta + \nabla u_\zeta$ be its cell gradient. Set $j_\eta = L_\chi e_\eta$, $p = B_\chi \eta$, and $\overline{F} = (\xi, R_\perp p)$. Module 11 gives $\mathrm{RC}\,W(\overline{F}) \ge 2\eta \cdot p$.

Setup 12.2. Lookup: $W(F) = \min(q_0(F), q_1(F))$ with $q_i(f,g) = f \cdot L_i f + g \cdot L_i^{\mathrm{d}} g$; these are the branches in Step 10.9. In Step 12.3, A, B are gradient matrices, whereas in Step 12.4 they are fields of such matrices. Strong L^2 convergence means $\|A_k - A\|_2 \to 0$; strong H^1 convergence additionally includes convergence of both weak derivatives. The constant K_0 bounds both branch matrices, as in Step 11.1. The two loads and detector are in Step 4.7–Step 4.8; e_ζ, j_ζ are the cell gradient and equilibrium flux of Step 1.6.

D1: Direct

Step 12.1. Bound one quasiconvex minorant

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 10.9.

For a smooth periodic φ and any quasiconvex minorant $h \le W$, show $h(F) \le \langle W(F + \nabla\varphi)\rangle$.

D1: Direct

Step 12.2. Take the supremum over minorants

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 12.1.

Deduce $\mathrm{QC}\,W(F) \le \langle W(F + \nabla\varphi)\rangle$ for every smooth periodic φ.

D2: Short proof

Step 12.3. Control the change in the obstacle

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 11.4, 10.9.

If the matrices of q_0, q_1 have norms at most K_0, prove $|W(A) - W(B)| \le K_0(|A| + |B|)|A - B|$.

D2: Short proof

Step 12.4. Pass energies through strong L^2 convergence

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 12.3.

If $A_k \to A$ in $L^2(Y)$, show $\langle W(A_k)\rangle \to \langle W(A)\rangle$.

D2: Short proof

Step 12.5. Allow H^1 periodic trials

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 12.2, 12.4.

Use Step 12.4 and periodic smoothing to extend the trial inequality to every $\varphi \in H^1_{\mathrm{per}}$.

D2: Short proof

Step 12.6. Make the joint field admissible

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 1.4, 1.9, 10.3, 10.4, 10.5.

Show $(e_\xi, R_\perp j_\eta) = \overline{F} + \nabla(u_\xi, v)$ for a periodic pair of H^1 potentials.

D1: Direct

Step 12.7. Identify the complementary energy pointwise

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 10.2.

Use $j_\eta = L_\chi e_\eta$ to show $j_\eta \cdot L_\chi^{-1} j_\eta = e_\eta \cdot L_\chi e_\eta$ almost everywhere.

D1: Direct

Step 12.8. Use one branch at each point

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 10.2, 10.9, 12.6.

For $e = (e_\xi, R_\perp j_\eta)$, prove $W(e) \le e_\xi \cdot L_\chi e_\xi + j_\eta \cdot L_\chi^{-1} j_\eta$ almost everywhere.

D2: Short proof

Step 12.9. Evaluate the averaged trial energy

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 1.10, 12.5, 12.6, 12.7, 12.8.

Deduce $\mathrm{QC}\, W(\overline{F}) \le 2\eta \cdot p - g_\chi$.

D1: Direct

Step 12.10. Finish the strict gap

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 11.10, 12.9.

Combine the last estimate with the Module 11 lower bound to prove $\mathrm{RC}\, W(\overline{F}) - \mathrm{QC}\, W(\overline{F}) \ge g_\chi > 0$.

D2: Short proof

Step 12.11. Check the order of choices

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 7.22, 8.18, 9.8, 9.9.

List the order in which ρ, N, the prescribed fraction θ, and its final periodic approximation are chosen in the composite proof.

D1: Direct

Step 12.12. Recognize the scope of the conclusion

Lookup: module 12 guide; setups 12.1, 12.2. *Uses:* 9.10, 9.11, 12.10.

Does this proof show that sixteen potentials are necessary, or that the two envelopes have been evaluated exactly? Explain briefly.

Optional oral checkpoints

Answer each question in a few sentences.

1. Why do two energy measurements exclude every laminate tensor limit?
2. Why use $PS^{-1}e_1$ and $PS^{-1}e_5$ as the horizontal loads?
3. Why does inverse closure help with lamination, while ordinary products can leave the matrix space during homogenization?
4. Explain the roles of four generators, eight auxiliary coordinates, sixteen original potentials, and thirty-two potentials in the energy problem.
5. Why is the energy difference fifth order even though the tensor itself has lower-order corrections?
6. What makes the path $a, b, b, b, -c$ useful, and how are other paths excluded?
7. Why does replacing χ_N by its average fail to justify binary recovery?
8. Why is the contrast fixed before the number of stripes is chosen?
9. Why is a limiting tensor at the right fraction not yet an actual periodic witness?
10. Why is the minimum of the original two gradient energies trivial?
11. What makes the minimum over $\mathcal{K}$ rank-one convex in this construction?
12. Which upper and lower bounds prove the strict envelope gap?

Background results and precise references

These are the general results supplied by the course. Each entry states the version used, identifies its applications, and points to a book section or a named theorem. The numbered steps work out their applications to the coupled system used in the construction.

B1. Periodic Sobolev spaces, completeness, Poincaré, and density

Input. A Hilbert space is a complete inner-product space. H^1 with its usual inner product is Hilbert, as are its closed linear subspaces. On the torus, smooth periodic functions are dense in H^1_{per}. For a zero-mean periodic v, $\|v\|_2 \le C_P\|\nabla v\|_2$. **Periodic version.** Extend periodically and convolve with a smooth approximate identity. Convolution preserves periodicity and converges in L^2 for the function and each weak derivative; subtract its mean if necessary. Poincaré also follows directly from Parseval: the zero Fourier mode is absent and $|k| \ge 1$ on every remaining mode. The finite number of potential columns causes no change: apply scalar statements columnwise. **Where used.** Module 1, Step 2.6, Step 8.14, and Step 12.5. **Reference.** Evans [4], Chapter 5, especially §§5.2–5.3 (Sobolev spaces and approximation), §5.5 (traces), and §5.8.1 (Poincaré inequalities). The preceding convolution and Fourier arguments explain the periodic adaptations; see also B6.

B2. Lax–Milgram

Input. If $\mathcal{H}$ is a real Hilbert space, a is a bounded bilinear form with $a(v,v) \ge c\|v\|^2_{\mathcal{H}}$ for $c > 0$, and b is a bounded linear functional, there is a unique $u \in \mathcal{H}$ with $a(u,v) = b(v)$ for all $v \in \mathcal{H}$. **Where used.** Step 1.1–Step 1.4; the bounds and the coercivity constant needed for this application are checked there. **Reference.** Evans [4], §6.2.1, the Lax–Milgram theorem.

B3. Symmetric matrices, operator norms, and order

Input. A real symmetric matrix has an orthonormal eigenbasis; its trace is the eigenvalue sum, its operator norm is the largest absolute eigenvalue, and its largest Rayleigh quotient is the largest eigenvalue. A positive-definite matrix has a positive square root and inverse square root, obtained by applying the scalar functions to that eigenbasis. For symmetric A, $\|A\| = \sup_{|z|=1} |z^\top A z|$. The identity $z^\top P^\top P z = |Pz|^2$ explains Gram matrices and rank. Congruence preserves order: $A \preceq B \Longrightarrow X^\top A X \preceq X^\top B X$. The last statement follows by evaluating the two forms at Xz. For an invertible matrix, $A^{-1} = \operatorname{adj}(A)/\det A$; thus a rational invertible matrix has a rational inverse. **Where used.** Modules 1, 3–5, 9–11. Order reversal under inversion is proved, rather than assumed, in Step 5.14. **Reference.** Horn–Johnson [6], Chapter 0 (basic matrix algebra and determinants), §2.5 (spectral theorem for normal matrices), §§4.1–4.2 (Hermitian matrices and variational characterizations), §5.6 (matrix norms), §§7.1–7.2 (positive definiteness), and §7.7 (matrix order).

B4. Separated scales, laminate composition, and the G-closure

Input. For uniformly coercive, bounded, self-adjoint quadratic gradient systems, first homogenize the fine periodic materials in each part of a coarser periodic pattern, then homogenize the coarse pattern. There are actual patterns in the original phases whose tensors converge to this iterated effective tensor. Fractions converge to the weighted fractions. The same statement holds for a finite hierarchy. With exact fraction correction it gives

$$C_i \in G_{\theta_i} \quad \Longrightarrow \quad \mathrm{Lam}_n(C_1, C_2; \lambda) \in G_{\lambda\theta_1 + (1-\lambda)\theta_2}.$$

Rational layer directions admit periodic patterns in a square cell; general directions follow by

approximation and continuity of the laminate formula. An aligned cell is used when deriving that formula in Module 2. **Scope.** The needed class consists of full-gradient quadratic systems with finitely many potential columns and a common coercivity constant. Milton's abstract field framework covers coupled systems; Allaire develops the locality and closure mechanism in conductivity and elasticity settings. **Where used.** Step 2.12, Step 9.8, and the interpretation of finite hierarchies. Step 9.4–Step 9.8 prove the exact-fraction correction from the weaker convergent-fraction recovery statement. **References.** Milton [10], Chapters 9, 12–13 and §§30.1–30.2; Allaire [2], §2.1.2, Theorem 2.1.2 and Lemmas 2.1.5–2.1.7 (locality, stability, and volume-fraction issues).

B5. Absolutely convergent Fourier series

Input. Summable coefficient sequences form a complete space under their ℓ^1 norm. For finite matrices use the operator norm in each summand. Absolute convergence permits rearrangement and gives a uniformly convergent Fourier series. Convolution estimates follow by the triangle inequality and $\|AB\| \le \|A\|\|B\|$; the order of matrix factors must be retained. These estimates are proved in Step 6.3–Step 6.7. Completeness of matrix-valued ℓ^1 follows by applying scalar completeness to each entry; all norms on a finite-dimensional matrix space are equivalent. **Where used.** Modules 6–7, including the geometric series and its differentiated versions. **References.** Katznelson [7], Chapter I, §6 (absolutely convergent Fourier series) and §9 (the higher-dimensional torus); Folland [5], Chapter 6, for completeness of L^p, applied to counting measure.

B6. Parseval, Fourier differentiation, and H^1 limits

Input. On $Y = (0, 2\pi)^2$ with normalized measure,

$$\|f\|_2^2 = \sum_{k \in \mathbb{Z}^2} |\widehat{f}(k)|^2.$$

For vector and matrix fields, use the Euclidean and Frobenius norms, respectively. A square-summable coefficient sequence is the Fourier sequence of an L^2 field. If $\sum_k (1 + |k|^2)|\widehat{v}(k)|^2 < \infty$, its finite Fourier truncations converge in H^1, and the weak derivative has coefficients $ik_j\widehat{v}(k)$. Conversely, testing a weak derivative against exponentials gives that coefficient relation. Finite sums verify differentiation first; their H^1 limit verifies it for the resulting function. For real fields, $\widehat{f}(-k) = \overline{\widehat{f}(k)}$. **Where used.** Step 6.10–Step 6.14, Module 7's frequency expansions, and Step 10.3–Step 10.5. **References.** Katznelson [7], Chapter I, §5 (square-summable Fourier series) and §9; Evans [4], §5.2, for weak derivatives. The explicit truncation argument is given in the answers to Step 6.12 and Step 10.5.

B7. The Cauchy determinant identity

Input. For distinct x_i and distinct y_j, with all $x_i + y_j \neq 0$,

$$\det\left[\frac{1}{x_i + y_j}\right]_{i,j=1}^{n} = \frac{\prod_{i<k}(x_k - x_i)\prod_{j<l}(y_l - y_j)}{\prod_{i,j}(x_i + y_j)}.$$

Where used. Step 7.19–Step 7.20, with $x = (0, 1, 4, 9)$ and $y = (1, 2, 3, 4)$. Pulling j out of column j accounts for the extra column factors in the course's determinant. **Reference.** Krattenthaler [8], §2.1, equation (2.7), the Cauchy double alternant. Its two Vandermonde factors may both be written with reversed differences without changing the sign.

B8. Lipschitz weak derivatives, traces, and distributions

Input. A Lipschitz function has bounded weak first derivatives agreeing with its derivatives almost everywhere. Its product with a smooth function obeys the weak product rule. The minimum of two Lipschitz functions is Lipschitz, since $\min(a,b) = (a+b-|a-b|)/2$ and absolute value is Lipschitz. On a cell divided into finitely many strips, Lipschitz formulas with matching traces have the piecewise weak derivatives: integration by parts cancels the interface terms. Distributional mixed partial derivatives commute, since both differentiate the smooth test function twice with the same result. **Where used.** Step 2.4, Step 8.5–Step 8.10, and Step 10.3. The answers carry out the interface cancellation and mixed-derivative tests. **References.** Evans [4], §5.2 (weak derivatives), §5.3 (approximation), §5.5 (traces), and §5.8.3 (Lipschitz functions); Folland [5], Chapter 9, for distributions.

B9. Uniform higher integrability for elliptic systems

Input. If A is periodic, measurable, self-adjoint, and $\mu I \preceq A \preceq MI$ with $0 < \mu \le M$ on $\mathbb{R}^{2\times m}$, there are $\delta > 0$ and $C < \infty$, depending only on the ellipticity bounds and the fixed dimensions, such that every periodic cell minimizer satisfies

$$\|F + \nabla u_{A,F}\|_{2+\delta} \le C|F|.$$

The constants are independent of phase geometry. This is the periodic form of local higher integrability for uniformly elliptic systems. **Bridge to this version.** Extend A periodically and apply the local system estimate to $U(x) = F^{\top}x + u_{A,F}(x)$ on a finite covering of the torus by balls with larger concentric balls in the periodic extension. Its gradient is the cell field, and the cell energy bound gives $\|F + \nabla u_{A,F}\|_2 \le \sqrt{M/\mu}\,|F|$. Periodicity controls the larger balls by finitely many cells. This yields the displayed estimate with constants uniform in A. **Where used.** Step 9.5–Step 9.8; Hölder's inequality then supplies the explicit modulus for small-set coefficient changes. **References.** Allaire [2], Theorem 1.3.41 and Remark 1.3.42 (the latter explicitly discusses elliptic systems); Meyers–Elcrat [9] for system higher-integrability theory.

B10. Quasiconvexity, rank-one convexity, and periodic testing

Input and hypotheses. Use finite real-valued, Borel measurable, locally bounded quasiconvex functions, with quasiconvexity defined by the Jensen inequality for zero-boundary $W^{1,\infty}$ tests. Such a function is rank-one convex and locally Lipschitz. For a Borel measurable, locally bounded integrand the usual quasiconvex inequality is equivalent to the periodic $W^{1,\infty}$ version. In particular the inequality holds for smooth periodic tests. Rescaling $\psi(x) = \varphi(2\pi x)/(2\pi)$ transfers the unit-cell assertion to $Y = (0,2\pi)^2$. In Module 12, first apply it to $h \le W$, then use the L^2 continuity of the obstacle W to allow H^1 trials (Step 12.3–Step 12.5). **Convention.** Texts often put the potential index in the first matrix coordinate. Transposing all matrices gives this course's spatial-first convention; it preserves the rank-one condition and the gradient Jensen inequality. **Where used.** Definitions in Modules 11–12, Step 12.1–Step 12.2, and the envelope comparison. **Reference.** Dacorogna [3], §5.2.1 (definitions), Theorem 5.3(i),(iv) (implication to rank-one convexity and local Lipschitz continuity), and §5.2.4, Proposition 5.13 (periodic characterization, including its local boundedness hypothesis).

B11. Integration and L^p estimates

Input. Cauchy–Schwarz is Hölder with exponents $2, 2$; Hölder with $p, q > 1$ and $p^{-1} + q^{-1} = 1$ gives $\langle |fg| \rangle \le \|f\|_p \|g\|_q$ for normalized measure. Fubini applies to the integrable functions used here, and dominated convergence applies under a common integrable bound. The graph of a continuous function $y = g(x)$ has area zero by its zero-length vertical sections. For a measurable $E \subset Y$, the function $a \mapsto |E \cap \{0 < x < a\}|$ is continuous: its increment is bounded by the area of the intervening vertical strip. Its endpoints are 0 and $|E|$, so any intermediate area can be selected. **Where used.** Energy estimates throughout; Step 8.5, Step 9.5, Step 9.7, and Step 12.4. **Reference.** Folland [5], Chapter 2 (Lebesgue integration, dominated convergence, Fubini) and Chapter 6 (L^p and Hölder inequalities). The slicing continuity needed in Step 9.7 also follows from the strip bound just given.

B12. Elementary compactness, continuity, and differentiation of series

Input. A closed bounded set in a finite-dimensional real space is compact. A continuous real function on a nonempty compact set attains its extrema. A continuous function on an interval takes all intermediate values. In finite dimensions all norms are equivalent. For a series of C^1 functions on a periodic cell, uniform convergence of the series of each first partial derivative, together with convergence of the function series at one point, permits differentiation term by term. The last fact follows by integrating the uniformly convergent derivative series along coordinate segments. In Step 6.19–Step 6.22 absolute majorants give the required uniform convergence directly. **Where used.** Step 4.3, Step 6.19–Step 6.22, Step 9.7, Step 11.2, and Step 11.11. **Reference.** Rudin [11], Chapters 2 (compactness), 4 (continuity), and 7 (sequences and series of functions, including differentiation).

References

[1] L. Cao, *Hierarchical Laminates Cannot Reproduce All Effective Properties of Planar Two-Phase Composites: A Counterexample with a 16-Component Potential*, *arXiv preprint*, 2026. arXiv:2610.37899. DOI: 10.48550/arXiv.2610.37899.

[2] G. Allaire, *Shape Optimization by the Homogenization Method*, Applied Mathematical Sciences 146, Springer, 2002. Publisher / DOI.

[3] B. Dacorogna, *Direct Methods in the Calculus of Variations*, 2nd ed., Applied Mathematical Sciences 78, Springer, 2008. Publisher / DOI.

[4] L. C. Evans, *Partial Differential Equations*, 2nd ed., Graduate Studies in Mathematics 19, American Mathematical Society, 2010. Publisher / DOI.

[5] G. B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Wiley, 1999. ISBN 9780471317166.

[6] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013. Publisher / DOI.

[7] Y. Katznelson, *An Introduction to Harmonic Analysis*, 3rd ed., Cambridge University Press, 2004. Publisher / DOI.

[8] C. Krattenthaler, "Advanced determinant calculus," *Séminaire Lotharingien de Combinatoire* 42 (1999), Article B42q, 67 pp. Journal text.

[9] N. G. Meyers and A. Elcrat, "Some results on regularity for solutions of non-linear elliptic systems and quasi-regular functions," *Duke Mathematical Journal* 42(1) (1975), 121–136. Journal / DOI.

[10] G. W. Milton, *The Theory of Composites*, Cambridge University Press, 2002. Publisher / DOI.

[11] W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw–Hill, 1976.